

CQ University Australia

CQU Rockhampton TAFE Consolidation

Hydraulic Specification

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CQU Rockhampton TAFE Consolidation Hydraulic Specification

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REV	DATE	DETAILS
A	12/02/2026	Issued for D&C Tender

	NAME	DATE	SIGNATURE
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Approved by:	Steven Mairs	12.02.26	SM

WSP acknowledges that every project we work on takes place on First Peoples lands.
We recognise Aboriginal and Torres Strait Islander Peoples as the first scientists and engineers and pay our respects to Elders past and present.

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1 Project Information

1.1 General

WSP Australia Pty Limited WSP has been engaged for the CQU TAFE Redevelopment Project as part of design team to deliver these specifications considering existing and future demands. Based on a 2022 masterplan study, the scope includes the provision of a new Trades Building (Building 92) and Horticulture Building, and the refurbishment of Buildings 19, 33, 34, 36 and Engineering (21A) Building.

As part of the initial scope of services, WSP attended the Rockhampton North and City Campuses to undertake a high-level review of the existing Electrical Services infrastructure reticulating around each site.

This specification takes into consideration existing items of plant and equipment, and its capacity for the proposed expansion.

The project delivery being design and construct, has been adopted for this project.

This Specification is to be read in conjunction with other tender documents issued by the Project Manager including:

- Principal's Project Requirements
- Conditions of Tender
- Conditions of Contract
- Contract Preliminaries

This Specification is a technical specification and for the purpose of this Specification, Works shall mean all works or work as described in this Specification and/or associated drawings

This technical specification and drawings are intended for a Design and Construct Contract and have been provided to communicate the design intent of the Works for the project and for the guidance of tenderers pricing the project. Notwithstanding this, the completion of the design, the co-ordination of the electrical services works, and achieving the system performance covered by this technical specification & Drawings shall form part of the Works.

The drawings and scope of works are based on information and drawings provided by the Client, other stakeholders and other design consultants which may contain conflicting data, incomplete design, coordination, or omissions. These layouts will be subject to change to suit final design. The contractor pricing the works shall make an assessment and make adequate contingency based on their experience with the procurement and risks involved in projects of this nature i.e., changes to requirements, site investigation outcomes, scope, and future design detail.

The Works requires the necessary resources to complete the detailed "For Construction" documentation and schedules, final equipment selections, testing and commissioning of the installation in accordance with the design intent and the requirements of this technical specification and drawings.

Notwithstanding this, achieving the system performance covered by this technical specification & drawings shall form part of the Works.

The work shall comprise that which is described in this document, any subsequent addendums.

The Drawings shall be read in conjunction with the respective existing services drawings and manuals and all associated documentation.

The tenderer shall visit the site to fully familiarise themselves with existing services installation and site conditions. Allowance shall be made to coordinate the existing services with the project staging and the scope of works covered under this Contract.

Include for all applicable site allowances and overtime required to meet the project programme.

Tender prices are to be submitted as a lump sum, fixed for the duration of the contract.

1.2 Project Team

The Project team includes the following team members:

PRINCIPAL/CLIENT	CQU
SUPERINTENDENT	TBC
PROJECT MANAGER	TBC
CONTRACTOR	The company who undertakes the Works
ARCHITECT	Thomson Adsett & Peddle Thorp
CONSULTANT ENGINEER (MECHANICAL, ELECTRICAL, VERTICAL TRANSPORTATION, FIRE PROTECTION, HYDRAULICS)	WSP Australia Pty Ltd
STRUCTURAL ENGINEER	WSP Australia Pty Ltd
FIRE SAFETY ENGINEER	WSP Australia Pty Ltd
SUSTAINABILITY CONSULTANT	WSP Australia Pty Ltd
ACOUSTIC ENGINEER	WSP Australia Pty Ltd
BUILDING SURVEYOR	TBC

1.2.1 The Project Description

1.2.1.1 General description

The project is the consolidation of the existing Rockhampton CQU and TAFE site. The project comprises of:

- A new Trades Centre Building
- A new Horticulture Building
- Refurbishment Building 19 level 3
- Refurbishment Building 33 levels 1 and 2
- Refurbishment Building 34 ground floor
- Refurbishment Building 36 levels 1 and 2
- Refurbishment of Engineering building
- Associated services enclosures/facilities

1.2.1.2 Utilities

The following utilities are relevant for this project:

Gas

Water – Existing site reticulation

Sewer – Existing site reticulation

Storm water

Fire Authority (QFD)

Telecommunications

Project Information

Electricity – Existing HV network is privately owned by CQU

2 General Requirements

2.1 General

2.1.1 Definitions

Unless the context otherwise requires, the following definitions apply:

1 Supply: 'Supply', 'furnish' and similar expressions mean 'supply only'.

Provide: 'Provide' and similar expressions mean 'supply, deliver and install' and include all testing, commissioning, tuning, interfaces etc.

Proprietary: 'Proprietary' means identifiable by naming manufacturer, supplier, Works Package, trade name, brand name, and catalogue or reference number.

Samples: Includes samples and prototypes.

Zinc-coated steel: Includes zinc-coated steel, zinc/iron alloy-coated steel and aluminium/zinc-coated steel.

2.1.2 Specification Objectives

The intent of this Specification is to:

1 Provide a basis for competitive tendering for suitably qualified tenderers.

Allow competent Contractors to price the works.

Provide a clear statement of technical and performance requirements against which compliance can be assessed.

Provide documentation required by Authorities to demonstrate design compliance with statutory requirements for the purpose of building permit application.

Provide documentation detailing the scope and quality of the project for the purpose of Client's confirmation of compliance to the Client's project brief.

Define the requirements of the Contractor with respect to Quality Assurance.

Define the scope of services including project management, installation engineering, testing commissioning, maintenance and documentation to be provided.

Specify the technical requirements of the Installation.

The Specification relies upon the Contractor having the necessary specialist services resources to complete the detailed design, testing and commissioning of the installation in accordance with the design intent and the requirements of the specification.

2.1.3 Design Intent

1 Intent: The Contract documents are intended to call for complete, operational systems. Provide all items (major and minor), equipment, accessories and incidental work required for the completion of the installation and to ensure full integration.

Diagrammatic layouts: Layouts of service lines, plant and equipment shown on the drawings are diagrammatic only, except where figured dimensions are provided or calculable. Before commencing work, inspect the site, obtain all necessary measurements and information.

2.1.4 *Discrepancies in Documents*

- 1 Give immediate notice in case of any ambiguity, discrepancy or inconsistency in the Contract document. Should conflict arise between the requirements of specification, drawings, mandatory requirements, standards including Standards Australia Codes, then the higher standard or most stringent requirements shall prevail and shall be deemed to have been included in the Contract. The Contract's order of documents precedence is not intended and shall not be applicable in this case.

Any ambiguities or discrepancies with the Design Requirements, Technical Requirements and / or Contract Drawings are to be highlighted at tender stage for clarification.

The Contractor is to review the tender documents, including other services, the architectural drawings, both general and detailed and satisfy themselves that the spatial allowances made for hydraulic services are adequate for the purposes of installing, testing and commissioning the works.

Any consequences of the Contractor's failure to advise the project manager, including any costs associated with relocation or modification to any area, shall be at the Contractor's risk.

2.1.5 *Referenced Documents/standards*

- 1 Use referenced documents/standards which are editions, with amendments, current 3 months before the closing date for tenders, except where other editions or amendments are required by Statutory Authorities.

2.2 Quality Assurance

- 1 Provide a Quality Assurance plan in order to propose, establish, maintain, monitor and document a quality assurance system covering all aspects of the design, purchase, fabrication, installation and completion of the works. The plan shall be in accordance with ISO 9001/.2 (as appropriate).

Designate a Project Quality Inspector to discharge the quality plan, which must include the following minimum:

- a Inspect the installation works regularly and provide installation defects reports on regular basis in order to systematically reduce the incidences of installation defects. Submit copy of such defects lists to the Consultant Engineer upon request. This task can be delegated to qualified site foreman if applicable.
- b Check and sign off all shop drawings (including all third-party shop drawings) for conformance to requirements prior to submission. All such documents shall list all non-conformances with the Specification and Drawings.
- c Check and sign off equipment compliance schedule against the particular Specification requirements and equipment schedules prior to submission. All such documents shall list all non-conformances with the Specification and Drawings.
- d Check and sign off all samples for conformance to requirements prior to submission. All such documents shall list all non-conformances with the Specification and Drawings.
- e Check all tests required for proper manufacturing of the equipment.
- f Check and sign off all manufactured items for compliance prior to dispatch to site.
- g Check and sign off installation of all items under this Specification. Provide signed off installation ITPs for every elements of installation (minimum floor by floor and/or for every individual area up to 1000sqm whichever is smaller).
- h Check all materials, welding, joining, terminations, fixing and finishes.
- i Check and sign off all associated ITPs for all on-site tests required to commission the works.
- j Check and sign off operating and maintenance manuals to ensure they contain adequate information to permit systems to be operated by the Client at the end of defects liability period. (Including adequate training and tuition of the nominated Client's representative).
- k Check and sign off all associated integrated ITPs for all integrated on-site tests required to commission the works and to be carried out by various systems.

General Requirements

All trades works including sub trades works shall form part of overall QA and required to have their ITPs, Shop drawings, technical submissions signed off also by the Project Quality Inspector. This sign off represent all required checking that these works are consistent and compliant with the rest of the works.

Submit the following documentation:

- l** Quality System third party certification, if any, to the Standards specified by the Joint Accreditation System of Australia and New Zealand.
- m** Quality manual detailing, corporate Q.A. policy statement, system element description, register of procedures and project specific ITPs.

Notice:

- a** Inspect witness points: If notice for inspection is to be given in respect of parts of the works, advise if and when those parts are to be concealed. Inspect and provide signed off installation ITPs indicating appropriate methods of installation been carried out.
- b** Inspection hold points: If notice of inspection is to be given in respect of parts of the works, do not conceal those parts without approval.
- c** Minimum notice for inspections to be made: 4 hours for inspections full time on-site, otherwise 2 working days for on-site inspections, and 5 working days for local pre-delivery inspections.
- d** Concealed services: Give notice so that inspection may be made of services to be concealed. Irrespective of Consultant Engineer inspection, inspect the works, sign off relevant ITPs and highlight any noncompliance and/or defects.

Witness tests: Give sufficient notice so that designated tests may be witnessed.

- a** Minimum notice for tests to be witnessed is 5 working days for site tests; and 10 working days for local pre-delivery tests.
- b** Test hold points: Do not carry out designated tests without approval.

Testing Authorities:

- a** General: Except for Site Tests, have type tests carried out by authorities accredited by NATA to test in the relevant field, or an approved organisation outside Australia recognised by NATA through a mutual recognition agreement. Co-operate as required with testing authorities.
- b** Site tests: Use instruments calibrated by authorities accredited by NATA.

Reports:

- a** General: Submit copies of test reports, including certificates for type tests, showing the observations and results of tests and compliance or non-compliance with requirements.
- b** Submit Installation Defects Reports carried out on regular basis to the Consultant Engineer upon request.

Concealment:

- a** If tests are to be carried out on parts of the work, do not conceal those parts and do not commence further work on those parts until the tests have been satisfactorily completed and compliance verified.

2.3 Electromagnetic Compatibility

2 Comply with Australian Communications Authority requirements for electrical and electronics products to limit electromagnetic interference (EMI).

- a** Passive products with C-tick or Regulatory Compliance Mark (RCM) to AS/NZS 4417 (Marking of Electrical and Electronic Products to Indicate Compliance with Regulations) and the Australian Electrical Equipment Safety Scheme (EESS).

3 Immunity:

- a Electrical and electronic apparatus: To AS/NZS 61000.6.1, Electromagnetic compatibility (EMC) – Generic immunity standard – Residential, commercial, and light industry, or
- b AS/NZS61000.6.2, Electromagnetic compatibility (EMC) – General standard – Immunity for industrial environments.

Harmonics and Voltage Surges:

- a Levels of emissions to be acceptable to the electricity distribution code. Keep the THDI current to 5% maximum for all individual components/equipment and at least to that level at the first supplying control panel for all components/equipment connected to the switchboard. Provide active harmonic filters within the component, or respective control panel. Do not rely on the external harmonic filters to be installed on the Main Switchboard(s).

2.4 Building Penetrations

2.4.1 *Fire resisting building elements*

- 1 Seal penetrations to comply with NCC using a system to AS 4072.1, Components for the protection of openings in fire-resistant separating elements – Services penetrations and control joints.

Comply with Section 3 of AS/NZS 1668:1, Fire and Smoke control in Multi Compartment Buildings.

AS4072.1 inter-alia Appendix B and AS1851 inter-alia Clause 17.2.3.1 shall be complied with in as-built documentation to identify all fire and smoke containment compartment walls and barriers. To satisfy specific identification requirements to AS4072.1 Appendix B4 and AS1851 Clause 17.2.4, permanently fixed labelling, tagging and signage of all passive fire and smoke containment systems must be provided for on-site identification. Fire stopping barriers and penetrations shall comply in particular with reference to a service label affixed adjacent to each fire stop barrier and service penetration (or close proximity group) as detailed on page 25 of AS4072.1 and page 165 of AS1851. This as-built provision amongst other as-built documentation and manuals is essential as part of handing over process of the project.

2.4.2 *Non-fire resisting building elements*

- 1 Seal penetrations around conduits and sleeves. Seal around cables within sleeves. If the building element is weatherproof, acoustic rated or subject to pressure, maintain the rating. Acoustically seal penetrations through plant room walls and floors.

2.4.3 *Chases*

- 1 Cut, drill and chase as necessary, where possible avoiding finished work. Where chasing of finished work is essential, make good all damage to the original standards of the finishes.

2.4.4 *Limitations*

- 1 General: Do not penetrate, or chase the following without approval:
 - a Structural building elements including external walls, core walls, fire walls, floor slabs, beams or columns.
 - b Acoustic barriers.
 - c Other building services.
 - d Membrane elements including damp-proof courses, waterproofing membranes and roof coverings.

Membranes: If approval is given to penetrate membranes, provide a waterproof seal between the membrane and the penetrating compound.

2.5 Required Submissions

- 1 Provide in good time to allow review without impediment to the programme, including possible amendment and resubmission:
 - a Copies of correspondence and notes of meetings with Authorities.
 - b Documents showing approval of the Authorities whose requirements may apply to the work.
 - c Certified schedule of compliance for all plant and equipment, prior to placing orders.
 - d Factory test results where applicable.
 - e All product data, performance test and commissioning results required by this Specification.
 - f Shop drawings for fabrication and installation of all equipment and items supplied.
 - g Manufacturer's product data for equipment, materials, components and systems including:
 - i Technical specifications and drawings.
 - ii Size, arrangement, operating and maintenance clearances.
 - iii Operating weight.
 - iv Type test reports.
 - v Performance and rating curves or tables marked with each selection showing for both duty point and for part load conditions the capacity; the energy consumption and power factor; and the sound power level.
 - vi Motor power requirements (both full load and starting).
 - vii Control details.
 - viii Recommendations for installation and maintenance.
 - ix Evidence of compliance with specified product certification schemes.
 - h Inspection, test and commissioning plans for every section of the works. Detail:
 - i The procedure of how to complete the task.
 - ii The skill or competency of the person undertaking the works.
 - iii The review or testing procedure to assure satisfactory completion of the task.
 - iv The person within the Works Package's organisation authorised to sign-off the task as accepted.
 - i Client handover/training proposals.
 - j Record and installation drawings: Record all changes to equipment and services layouts, wiring and any other items during the construction period, which may have been incorporated into these works.
 - k Operating and Maintenance manuals.
 - l Maintenance/service records during the defects liability period.

Give notice before commencing work affected by the submissions, unless the submissions have been reviewed with no exception taken.

Do not commence production of equipment, materials, components of equipment, components or systems until submission has been reviewed with no exception taken.

Maintain and submit monthly a schedule of shop drawings and submissions to identify all proposed drawings and submissions designated in the following categories:

- a Accepted.
- b Submitted, awaiting acceptance (with submission date and revision number).
- c Not yet submitted.

Submit electronically (PDF or alternative agreed format).

Identify the project, Works Package, supplier/manufacturer, applicable product name, product number, included product options, and relevant specification references.

Where selected equipment or systems do not fully comply with this Specification and associated documents, submit details of proposed deviations for review. Where such deviations require any changes to the structure, building works or

services, these shall be provided by the Contractor without variation to the Contract. Apart from deviations listed, it will be taken that the materials and equipment proposed meets with all other specified requirements. Review of equipment and materials is undertaken on this basis, and that non-complying equipment/materials may be rejected at any time – even if already installed.

Submit amended documents as appropriate, highlighting changes since the previous submission.

2.5.1 *Samples*

2.5.1.1 *Timing*

- 1 Co-ordinate submissions of samples as listed in the Schedules. Do not cause delays by making late submissions or submitting inadequate samples.

2.5.1.2 *Quantity*

- 1 Submit a sample of each designated item and 2 copies of supporting documentation. Include ancillary items such as fasteners and mounting brackets. Amend and resubmit samples which are not accepted.

2.5.1.3 *Identification*

- 1 Identify the project, Contractor, their agents or supplier, manufacturer, applicable product, model number and options, as appropriate and include pertinent Specification references.

Include service connection requirements and product certification.

Identify non-compliances with project requirements, and characteristics that may be detrimental to successful performance of the completed work.

2.5.1.4 *Sample acceptance and retention*

- 1 Do not commence work affected by samples until the samples have been accepted. Submit further samples as necessary.

Keep approved samples in good condition on site, until Practical Completion.

Incorporate in the works samples, which have been approved for incorporation. Do not incorporate other samples.

Installed items will match accepted samples throughout the works.

2.6 *Materials, Equipment and Components*

2.6.1 *General*

- 1 Provide new materials, equipment and components shall be new, of good quality and fit for purpose, selected for a reasonable service life. Do not provide without approval products that are obsolete, discontinued or about to be discontinued.

Identification of a proprietary item does not necessarily imply exclusive preference for the item so identified, but indicates a deemed to comply item.

Manufacturers: Provide equipment and associated accessories which are the products of established manufacturers regularly engaged in the manufacture of such equipment, who issue comprehensive rating data and certified test data on their products.

Consistency: For the whole quantity of each material or product use the same manufacturer or source and provide consistent type, size, quality and appearance.

General Requirements

Safety: Provide all necessary safety devices for the protection of personnel against injury and the protection of plant and equipment against damage including relief valves, belt guards, safety railing, effective earthing of electrical components, electrical interlocks, warning lights, and signs, alarms and local lighting. Provide permanent lifting eyes for equipment exceeding 40kg.

General: Select, if no selection is given, and transport, deliver, store, handle, protect, finish, adjust, prepare for use, and use manufactured items in accordance with the current written recommendations and instructions of the manufacturer or supplier.

Instructions: Submit the recommendations and instructions, and advise of conflicts with other requirements.

Project modifications: Advise of activities that supplement, or are contrary to, manufacturers or suppliers' written recommendations and instructions.

Product certification: If products must comply with product certification schemes, use them in accordance with the certification requirements.

Provide permanently fixed plates indicating manufacturer, model, serial number, capacity and electrical data for all equipment. Lettering height: 5mm maximum, 1.5mm minimum.

Provide materials or products are supplied by the manufacturer in closed or sealed containers

2.6.2 *Finishes and Visual Elements*

1 Surfaces shall be flat and free of noticeable distortion, warping and twisting.

Edges and corners accessible to users dressed to avoid injury if touched.

Pressings shall be accurate in dimension and profile to ensure consistent mating and clearance gaps and pressing shape.

Mating surfaces between adjacent pressings - flush and accurate to produce smooth distortion free joint.

Provide protection against damage until Practical Completion.

Surface shall be clean, undamaged and free of stains

2.6.3 *Manufacturers' or Suppliers' Recommendations*

1 General: Select, if no selection is given, and transport, deliver, store, handle, protect, finish, adjust, prepare for use, and use manufactured items in accordance with the current written recommendations and instructions of the manufacturer or supplier.

Instructions: Submit the recommendations and instructions, and advise of conflicts with other requirements.

Project modifications: Advise of activities that supplement, or are contrary to, manufacturers or suppliers' written recommendations and instructions.

Product certification: If products must comply with product certification schemes, use them in accordance with the certification requirements.

2.6.4 *Cleaning and Protection of Finished Work*

During handling and installation of work at the site, clean and protect work in progress and adjoining work. Apply suitable protective covering on newly installed work where required to ensure freedom from damage or deterioration at a time of Completion; otherwise, clean and perform maintenance on newly installed work as frequently as necessary throughout remainder of construction period. Adjust and lubricate operable components to ensure operation without damaging effects.

2.6.5 *Prohibited Materials*

1 Do not use any of the following materials:

- a Lead where the metal or its corrosion products may be directly ingested, inhaled or absorbed, or any lead-based paints or primers.
 - b Chromate paint pigment or chromate water treatment.
 - c Urea formaldehyde foam or materials which may release formaldehyde in quantities which may be hazardous or irritant.
 - d Synthetic mineral fibres except in accordance with Worksafe Australia National Standard and National Code of Practice.
 - e Materials in which chlorofluorocarbons, hydrochlorofluorocarbons or hexa-fluoroacetones have been used as a blowing agent.
 - f Chlorofluorocarbons.
 - g Polychlorinated biphenyls.
 - h Other substances generally known to be deleterious to health or safety or which would adversely affect the works.
 - i Any that would be contrary to achievement of a required project Green Star or other sustainability rating.
-

2.7 Installation

2.7.1 General

- 1 General: Carry out the work in a proper and workmanlike manner.

Arrangement: Install equipment and services parallel or perpendicular to building elements. Organise reticulated services neatly. Provide for movement in both structure and service. Under suspended ground floors, keep services at least 150mm clear above ground surface, additional to insulation.

Movement and expansion: Provide expansion facilities in ductwork, piping, cables, cable trays and supports to accommodate thermal expansion and movement at structural expansion joints.

Protection: Protect equipment from weather and the ingress of dirt, moisture, vandalism and tampering.

Access: Provide access to all components requiring entry, inspection or maintenance.

If interruptions to supply of any service are required co-ordinate the shutdowns to the satisfaction of the Superintendent, advise the occupants, minimise inconvenience, and advise when supply is reinstated. Arrange shutdowns out of hours.

2.7.2 Services/utilities connections

- 1 If the utility provider elects to perform or supply part of the works, make the necessary arrangements. Install equipment supplied, but not installed, by the utility provider.

Connect to utility provider services or service points. Excavate to locate and expose connection points. On completion reinstate the surfaces and facilities which have been disturbed. Pay connection charges.

2.7.3 System integration

- 1 Interconnect system elements so that the installations perform their designated functions.

2.7.4 Setting out openings and making good

- 1 The Specification Drawings are diagrammatic only and unless otherwise stated shall not be used for determining the precise positions of equipment outlets and like features. The exact location of these shall be determined on site and/or from shop drawings.

General Requirements

All chases, ducts, recesses and penetrations in structural elements not shown on the building and structural drawings shall be subject to approval.

Provide the precise location of all openings, fixings and similar items of work required for these works.

Determine requirements at such times as not to cause delay to the work.

Check all items to ensure correct positioning.

Wrongly located or omitted openings, fixings and similar items of work resulting from incorrect or lack of inter trade coordination shall be remedied and paid for under these works.

2.7.5 *Works programme and time chart*

- 1 Sufficient information of construction phase activities shall be provided to enable preparation of a detailed construction programme incorporating activities of every trades.

The programme of work shall be arranged in consultation with all Trades and the installation shall be completed according to the agreed programme.

The Works programme shall clearly show the start and completion dates with the typical activities listed below. The list is indicative.

Submission of shop drawings and major technical submissions.

Shop fabrication

Ordering of plant and materials.

Delivery of plant and materials.

Erection, testing and painting/identification of pipework/cable trays/ conduits/ducts in false ceilings.

Erection of pipe risers and installation of electrical wiring in vertical building risers before building riser walls are completed and closed.

Permanent power connection.

Commencement of fitting offs and final terminations.

Authorities acceptance tests.

Commencement of testing.

Commencement and Completion of plant commissioning.

2.7.6 *Metalwork*

- 1 General: Use metalwork capable of transmitting the loads imposed, and sufficient to ensure the rigidity of the assembly without causing deflection or distortion of finished surfaces. Construct to prevent rattle and nuisance.

Metal separation: Prevent contact between electrolytically dissimilar metals, by using concealed insertion layers.

Edges and surfaces: Keep clean, neat and free from burrs and indentations. Remove sharp edges.

2.7.7 *Fixing*

- 1 Fix all plant to structure in approved manner, either directly or via secondary structure as required. Submit details of types of fixings, locations and loads for approval.

Fix only light weight items to non-structural building elements.

Do not pierce waterproof roofs, floors or walls with fixings.

Fasteners to comply with appropriate Australian / New Zealand Standards.

General Requirements

Use proprietary corrosion resistant fasteners capable of transmitting the loads imposed, and sufficient to ensure the rigidity of the assembly. Use metal expansion bolts for concrete and masonry. Do not use explosive charge fixings without approval.

Corrosion protection: Use corrosion resistant, chemically and electrolytically compatible fasteners. Provide insulating spacers where necessary.

Where pipework supports are installed for water and drainage services within server and comms rooms, supports are to be stainless steel or equivalent of a non galvanized finish to eliminate 'zinc whisker' formation.

2.7.8 *Installation in Communications or Electrical rooms*

- 1 Drainage installations to have safe tray located directly under all drainage to prevent any future leakage from system discharging onto electrical or communications equipment.
- 2 Safe trays shall discharge to atmosphere in the same room in order to easily identify that leakage is occurring and rectification.

2.8 Co-ordination

- 1 Comprehensive co-ordination of all services and associated building work form part of the Works.

These responsibilities shall include:

- a Co-ordination of the works as required.
- b Management of the advance procurement of equipment to meet programme.
- c Co-ordination of in-ground and in-ceiling services routes, including trenching, trays and pipework.
- d Management of the preparation and approval of fully co-ordinated drawings including penetration drawings.
- e Management of services testing and commissioning.
- f Ensuring provision of all test results and required certifications to obtain Certificate of Occupancy from the Building Surveyor.

During the construction period the works require that a competent supervisor is provided on site responsible for the comprehensive co-ordination of all engineering services, including civil, structural, electrical, mechanical, hydraulics, fire and lift services.

Responsibility for the proper direction, supervision, control and co-ordination of the work forms part of the works and shall assist all other trades in every way possible in meeting this responsibility.

All services shall be thoroughly co-ordinated prior to installation. Should work proceed without co-ordination being undertaken and authorisation by those with such responsibility and should other trades subsequently not be able to effectively proceed with their respective installation, the work initially undertaken shall be removed and reinstalled at the expense of the Contractor.

Any claims for additional costs or delays due to the lack of effective co-ordination by the various trades will be rejected.

Arrange and co-ordinate all interconnection testing with other services systems, e.g. fire mode tests, power failure testing etc.

2.8.1 *Co-ordination of installation with ceiling tiles and finishes*

- 1 Co-ordination shall be carried out to ensure that all building and services elements such as air ducts, outlets and light fittings, sprinklers, speakers and other ceiling elements are installed in a co-ordinated manner, such that all services can be connected and maintained as indicated on the Architectural Reflected Ceiling Plans.

2.9 Safety in Design

- 1 Safety in Design risk management shall form an integral part of these works and shall conform to the requirements of the Workplace Health and Safety (WHS) Act 2011 & WHS Regulations 2011.

The Safety in Design process shall conform to the State-relevant Compliance and Guidance documentation.

General principles of risk management to be followed shall be as AS/NZS ISO 31000 (2009) Risk Management – Principles and Guidelines.

The National Standard for Construction Work (NOHSC:1016) shall be used to align design risk management practice with construction risk management practice.

The Safety in Design Risk Assessment shall include but not be limited to:

- a Understand the range of work activities associated with the intended use of the building / structure as a work place.
- b Identify hazards, assess risks (quantify and rank), identify control measures, implement control measures, review and prepare risk register prior to construction of work. Monitor and review throughout construction and operation. The process includes review and acknowledgement of residual risks identified and communicated by the designer as defined by the WHS Act and Regulations.
- c Elimination or where this is not reasonable practicable, minimisation of any risk to health and safety of any person. The process includes, recommendation of design alternatives that will eliminate / reduce risks, recommendation of control measures for residual risks; organising participate and facilitating Safety in Design workshops as required.
- d Report in writing, on health and safety aspects and risks of the design identified. Implementation of control measures which shall be monitored and reviewed throughout construction and operation.
- e Provide at the conclusion of the project all residual risk information, in the form of a Safety in Design File and the related instruction and training as required to ensure an understanding of the safety aspects of the installation.

Where local regulatory practice differs from the above the more onerous shall be used as the basis for compliance.

2.10 Existing Site

- 1 With the submission of their tender the Contractor shall be deemed to have familiarised themselves with any as-built documentation, visited the site to ascertain any latent conditions and be fully conversant with the conditions and restrictions pertaining to, including but not limited to any local Authority requirements, in order to be able to provide the all items relating to the works.

3 Project Requirements and Description of Works

3.1 General

- 1 Unless varied by Specification Addenda, the requirements set out below in this section are mandatory and no deviation is permitted.

List and allow in your tender form all of the items called in and listed in this section. No deviation from requirements of this section is permitted unless approved by Superintendent in writings during the tender process.

3.2 Roles and Responsibilities

3.2.1 *Obligations of the contractor*

- 1 In writing this Technical Specification the Client expects and relies upon the Contractor to provide its own personnel and its trades possessing specialist trade expertise necessary to complete the works in accordance with this Technical Specification which form part of the overall Contract documentation.

In addition, the Contractor has the following obligations:

- a To complete the design while demonstrating compliance with the design intent.
- b To raise in good time, issues requiring design input or clarification from the Consultant Engineer, particularly in respect to:
 - i interpretation of the Specification or drawings;
 - ii problems in complying with the Specification together with suggested alternative/substitutions;
 - iii matters in the Contractor's opinion, are omissions not discovered during the tender process, together with suggested alternatives/substitutions
- c To certify compliance with Authority requirements.
- d To pay all fees applicable to the works.
- e To obtain all Authority permits and certificates to allow the progress of the work.
- f To provide manufacturer's and construction drawings of sufficient detail to allow proper fabrication, co-ordination and installation and a scale not less than 1:50; incorporating all post-tender changes to architectural and structural drawings.
- g To provide samples and prototypes where specified.
- h To provide manufacturer's factory test records (type tests and witnessed tests) and site installation inspection and sign-off certifying that the completed installation comply with the manufacturer's installation guidelines and requirements
- i To provide the completed Services installation that is fully and correctly commissioned and fine-tuned and, that is functional and operable efficiently in compliance with all requirements of this Technical Specification.
- j To provide copies of all monthly/ quarterly maintenance and servicing reports required to the Consultant Engineer.
- k To certify compliance with Contract documents, including all variation instructions, at Practical Completion
- l To implement procedures to ensure that only competent tradesmen are used for the works.
- m To be cognisant of the role of all relevant parties during the construction phase of the project and to assist them in the conduct of their duties wherever possible.
- n To contribute, in the spirit of partnering, towards the successful execution of the contract.

3.2.2 *Contractor's design responsibilities*

Specifically, for specialist services trades, the Contractor shall be responsible for the detailed design activities listed below, as these activities are considered to be normally undertaken through the custom and practice of the industry.

- 1 The Contractor shall be responsible for ensuring that the detailed design undertaken is fully co-ordinated and compatible with the remainder of the project design.

Check space requirements of equipment and services which are indicated diagrammatically in the Contract documents. Select equipment with dimensions to suit the available space.

Lay out equipment and services to be accessible for operation, maintenance and replacement and so as not to interfere with access to other installations. Make offsets as necessary.

Set out access ways 2.1m high and 1.0m wide (minimum) to all major plant clear of all obstructions, unless otherwise approved.

Co-ordinate the layout of plant and services with the building layout and structure, and with other plant and services.

Neatly group services, with separate layers for crossing services.

Interface details with other trades.

Size and location of penetrations in walls and floors.

Physical co-ordination of installation with other trades.

Provision of electrical loads to electrical trade and compare with design loads cable sizes and circuit protection devices, seeking design direction where discrepancies occur.

Co-ordination of the construction of the installation.

Details of electrical wiring and control diagrams of all equipment supplied showing all interconnections between equipment to enable the necessary wiring to be undertaken.

Thermal expansion accommodation and anchorage, including provision of bellows or bends, taking into account final installation details and consistent with specified requirements.

Mounting, support and fixing details, and fasteners, including any secondary structure which may be required.

Cable installation details and derating factors.

Settings for protection equipment, time delays, time switches etc.

Capacity, location, design and sizing of cable support, trunking and conduit systems.

Acoustic design or modification of actual selected equipment that may require change, to meet with the noise levels specified. Such change to approval of the Consultant Engineer. Specified levels to be achieved with all plant operating.

Selection of all anti-vibration mountings to suit the particular application of the mounts

Valve, damper and access locations.

System water capacities and chemical additives.

Final locations of control sensors, detectors, thermostats, switches, outlets.

Calculate system resistances to fluid flow based upon the actual plant layout and selected equipment. Allow for dirty filters and diversified flows.

- 2 Provide detailed design, certification, obtain necessary authority approvals and licences for the sewer treatment plant and the water treatment plant.
- 3 Provide detailed design, certification for all sanitary drainage to meet the performance criteria noted within this specification and documentation. Shop drawings to be submitted with clear detail ensuring all performance requirements are met including but not limited to:

- a Valving arrangements.
- b Chamber material, size and depth.
- c Anti-buoyancy measures.
- d Access lid sizes and class rating.
- e Pump detail and specification.
- f Pump removal methodology and arrangement.
- g Pump switching levels and switching signal type.
- h Control panel details.
- i Inlet/outlet pipework arrangement and levels.
- j Emergency storage volume/time catered for.
- k Rising main size.
- l Rising main air bleed stations and requirements over entire length of rising main.

3.2.3 *Alternative products*

Should the part of the work be proposed to be done quicker, better or more effectively by substitution of materials or methods other than those specified, the details of such substitutions or alternatives shall be included in the Tender, including a comprehensive Report including any performance and life cycle analysis in support of the proposals.

Where the words 'equal or approved' are used in this Technical Specification, permission may be requested to use a substitute for what is specified, provided it and the relevant manufacturer of the item certify in writing that the substitute is of equal, or better quality and effectiveness to that specified.

Submit for approval full details to allow verification that the alternative products comply with the Contract Documents. Provide a certificate for each alternate product confirming that the proposed alternative product complies with the contract documents. Include all compliance and non-compliance items related to Project Brief, Specifications and Contract Drawings. State if use of proposed alternatives will necessitate alteration to other parts of the works and include consequential costs.

- 1 Allow for additional work resulting from the utilisation of an approved alternative product, including additional or revised statutory approvals, changes to adjacent work, re-submission of shop drawings and any costs incurred by the Consultant Engineers in assessing such alternative products. Allow for Consultant Engineer Review cost based on agreed Market hourly rates. These sums are to be paid irrespective of the alternatives being selected or not.

Proposals for substitution must include details of:

- 1 Changes to adjacent work, if any.

Cost differences.

Life cycle cost differences

Quality differences and advantage to the Client.

Lead time and program changes.

Certificate of compliance.

3.3 Existing Services

Obtain full information regarding all existing services, including, but not limited to, water, gas, sewer, stormwater, electrical and telephone - wherever new piping is to be installed.

Pay all costs involved in making good any service damage while excavating, installing or backfilling for any service noted herein. Investigate for exact location, depth and size of all existing services prior to any installation being carried out.

Existing services shall be maintained and operational until new works are installed. Allow tracing services as necessary to determine if services shown or found are 'live' and performing connection to new services with the minimum of downtime. All disconnections and shutdowns shall be approved prior to work being commenced.

Where existing branch lines are located and are not shown on the drawings or 'As Constructed' drawing (available at request from the Superintendent) allow to trace and make connection to new services or cap off as appropriate.

Existing services shown on drawings have been documented based upon existing records. The accuracy of location and size etc shall be verified by the contractor prior to commencement of works.

3.4 Staging or Sequencing of Works

Allow to stage works as directed by the superintendant and/or project manager. Allow for submission of plumbing approvals to match any separable portion handovers so as not to impede the certificate of classification process.

Existing services shall be maintained and kept operational until new works are installed. Allow for all temporary services to maintain continuity of services throughout the project as necessary

3.5 Description of Works

Whether or not particular works are described in the Technical Specification all items and materials needed for the complete works are required and shall be installed unless clearly excluded.

The project includes the detailed design, supply, delivery, installation, commissioning, tuning, testing, placing into service, maintenance, warranty and defects liability of materials, labour and plant of the services systems in accordance with this Technical Specification. All components and systems shall be complete in every respect and tested and commissioned unless otherwise noted specifically.

3.5.1 *General*

- 1 Allow to decommission, make safe and remove any existing redundant services to be deleted within the area of works.

Site visit and familiarisation with the site conditions and scope of work required under this contract and making required allowances to complete the works.

All necessary negotiations including formal submissions as required with all Authorities/utilities having jurisdiction including obtaining of design approval and final certification of the installed systems by an independent certifier approved by the Authority. Payment of all fees associated with the authorities/utilities connections and permits for the installation

Provision of detailed safety in design analysis and ensure the installation is in compliance with the Site's safety in design proposal including all regulatory requirements.

Engagement and Coordination with the Green Star / Sustainability Consultant and the Independent Commissioning Agent, including timely preparation of documents as required by them

Allow for all requirements in relation to alternatives proposed for the project. Allow for Consultant Engineer review cost based on agreed market hourly rates. These sums are to be paid irrespective of the alternatives being selected or not.

Collaboration and working with Mechanical Trade who is responsible for the management and coordination of overall services shop drawing process including providing timely input into the process to allow fully coordinated shop drawing production for all services Active engagement and provision of nominated staff for input into the multidisciplinary

coordination workshops in relation to shop drawing production as well as onsite installation methodology and integrated commissioning of multi-disciplinary interfaces

Coordinate with the Electrical trade and advise final equipment selections and their power requirements prior to their ordering and installation of submains to respective switchboard to ensure correct submain cable selections- Submains cabling from the MSB is provided to the primary mechanical/Hydraulic/Lift switchboards. Provide all lugs, cable glands and gland plates for the Electrical trades so they can terminate the cabling and do the final connection to your respective switchboard. Any submains/sub circuit to secondary switchboards/control panels shall be included in the respective services (Mechanical, Hydraulics, Fire, Civil, Vertical Transportation) works

Provision of QA plan including inspection, testing, commissioning ITPs for all systems installed

Preparation of “For Construction” and “Shop Drawings” coordinated with other services prior to commencement of works including services layouts at the same scale as the design drawings.

Provision of all samples and technical submissions. Samples include;

- a Basins
- b Water Closets
- c Urinals
- d Tapware
- e Longitudinal Trench grates.
- f Rainwater outlets
- g Inspection point covers
- h Floor grates
- i Acoustic and Thermal insulation along with appropriate ESD certification.
- j In wall tundishes

Identification and coordination of penetrations, provision of fire/smoke sleeves and sealing of all wall, floor, ceiling penetrations to the required fire/smoke/acoustic rating.

Provision of vibration isolation for all rotating equipment.

Provision of seismic restraints, including design.

Provision of all painting and finishes including identification and labelling schedules,

Provision of As-Built drawings and O&M Manuals in compliance with a uniform overall Site Services Documentation System.

Provision of all training for staff in the operation and preventative maintenance of the systems. . Training of the nominated building facility management staff to the level of understanding of day by day operation of the services systems

Twelve (12) months statutory testing, servicing, maintenance, defect liability and warranty of all services systems/equipment installed under this contract from the date of Practical Completion

Fine tuning commissioning of all systems during the defects liability period as per Green Star/NABERS requirements. The tuning shall involve inspection, check and all adjustments of all systems and controls every three (3) months after the date of Practical Completion. Detailed final recommissioning of all systems shall be carried out at the end of the defect liability period.

3.5.2 *Inter services works*

- 1 Provision of open end tundishes for all mechanical equipment waste. Tundishes shall comply with “Materials” section.

Provision of water supply to mechanical plant. Terminate with valved branch. Extension and final termination to mechanical plant by Mechanical Services Works Package.

Provision all downstream (of the hydraulic distribution board or control panel) power and control wiring associated with each hydraulic services.

Provision of volt free terminal strip in Control Panel/equipment for generating status of equipment on BMS.

3.5.3 *Hydraulic services works*

- 1 Provision of a complete sewer drainage system including connection to the sewer mains connection, pipes, junctions, bends, pits, floor wastes, excavation, supports, backfilling, testing and sundry equipment required for the installation. Coordination shall be carried out prior to any setting out, excavation and pipe installation taking place.

Provision of a complete stormwater drainage system including connection to stormwater connection provisions and all necessary pipes, junctions, bends, pits, floor wastes, excavation, supports, backfilling, testing and sundry equipment required for the installation.

Supply and fix the taps and faucets specified in the “Materials” section or as selected by the Architect.

Provision of a complete domestic cold water system including all fixtures, fittings and faucets requiring domestic cold water. Include for all pipework, bends, offsets, brackets, pumps, taps and faucets and sundry equipment required for the installation.

Provision of a complete domestic hot water services system including all piping, fittings, supports, insulation, hot water heaters, valves, circulating pumps and other sundry items of equipment required for the installation. The hot water system shall consist of a flow and return system from the hot water plant where documented. The hot water is to be delivered at 65°C with electric booster hot water heaters in each commercial kitchen area to boost the hot water to 82°C (if documented). Thermostatic mixing valves and tempering valves will be located in all other areas to maintain the water at 50°C or 42°C.

Supply and install a hot water circulating pump to the requirements as specified in the “Pumps” section.

Supply and install thermostatic mixing valves to all personal hygiene fixtures, as indicated on the drawings, to manufacturers requirements and in accordance with the local authorities’ requirements. Valves shall be set to 42°C.

Provision of a complete domestic hot water services system including all piping, fittings, supports, insulation, hot water heaters, valves and other sundry items of equipment required for the installation. The hot water system shall consist of an instantaneous point of use system. The hot water is to deliver at 50°C or 60°C, dependent on area of use. Thermostatic mixing valves and tempering valves will be in all other areas to maintain the water at 50°C or 42°C.

Provision of a complete gas reticulation system including mains connections, meters, regulators, filters, pipework and sundry items to comply to the local gas authorities’ regulations. Provide a system of piping to supply medium / low pressure (7.0 / 3.0 kPa) natural gas from the authority meter location, and reticulate through the building to appliances nominated in the drawings.

Arrange for the local Gas Authority to supply and install an appropriate natural gas meter assembly complete with an automatic shutdown valve.

Supply and install an appropriate Trade Waste treatment device to adequately treat Trade Waste Drainage prior to discharge to the sanitary drainage and/or sewer network in accordance with local authority requirements and the NCC.

Provision of a complete the rainwater harvesting plant system and recycled water service including all piping, fittings, supports, filters, treatment devices, valves, pumps and other sundry items of equipment required for the installation including provisions for interface to BMS. The harvested rainwater is to be circulated through the building and a flushing device for the main runs must be provided to prevent dead legs. Provide a main’s water diversion to ensure sufficient water to all fixtures being supplied by the rainwater harvesting plant at all times. The main’s water diversion supply has to be metered separately. This system must be capable of indicating water levels in the tank and control the tank diversion with main’s water.

Site services infrastructure:

- a Sewer: Connect new 100mm sanitary drainage to nearest live and maintained existing sanitary drainage stack/sewer main/manhole for each area of works.
- b Water: A new water connection shall be made to the nearest existing live water main for each building.
- c Stormwater: New stormwater connections shall be made to the nearest live existing/new stormwater network adjacent to each building as required.
- d Gas: A new connection shall be made to the existing authority gas main located adjacent building 36.

3.6 Shop Drawing and Construction Drawing Requirements

- 1 Prepare and submit dimensioned drawings showing details of the fabrication, layout and installation of all plant and equipment, including relationship to building structure and other services.

Prepare and submit drawings of penetrations and “built-in” components in the same form and manner as described for the processing of shop drawings. All penetration measures shall relate to grid lines, co-ordinates or relative levels.

Submit co-ordinated penetration drawings in sufficient time for review. Any failure to submit requirements within sufficient time for revision of structural drawings, resulting in re-working of the structure or re-ordering of structural components shall be responsibility of Contractor and shall not constitute any claim for variation or delay. All penetration measures shall relate to grid lines, co-ordinates or relative levels.

Provide drawings generally in accordance with the design drawings supplied with this specification.

Sheet sizes: Standard metric series, all sheets the same size. Plant room layout plans 1:50 scale, with sections and details at 1:20 scale. All services Floor plans at 1:100 scale, with detailed sections.

Drawings:

- a Access to and removal of all plant including locations and sizes of access doors to be installed in finished building surfaces.
- b Schematic and layout drawings of piping, tray and conduits, electrical and controls.
- c Switchboard and control panel layouts and schematics with terminal numbering.
- d Locations of all control sensors, valves and actuators.
- e Detail of all interfaces with other services and utilities.
- f Services co-ordination drawings for spatial co-ordination with building structure and other services with details for restricted locations.
- g Control logic diagrams with settings.
- h Foundations, plinths, chases, ducts, pits and penetrations through structure.
- i Layout and details of services cast in concrete. (for co-ordination and review by structural engineer /architect)
- j Structural support details and methods of fixing to structure.
- k Construction loadings (for structural engineer review).
- l Lifting point loads. (for structural engineer review)
- m Maintain current sets of drawings on site and progressively record variations which lead to creation of as-built drawings.
- n Detailed drawings for all tanks and pump stations clearly demonstrating compliance with the performance requirements nominated within this specification and documentation as well as compliance with all relevant Australian Standards.

Please note when shop drawings review comments states “satisfactory subject to comments”, resubmit the revised drawings with comments incorporated within a period of no longer than 2 weeks.

3.7 BIM/Revit Requirements

- 1 The project is documented in a 3D authoring tool to produce the Consultant Engineers Model(s). The Consultant Engineers Model(s) forms the basis of the creation of Consultant Engineers Drawings (2D) which will be considered

the main form of contract documentation together with this Specifications for Tendering purposes. Where considered appropriate, the communication of design intent via the Consultant Engineers Drawings have been prioritised over the detailed modelling of some elements in 3D.

Co-ordination of model elements in the Consultant Engineers Model(s) has been completed to a degree adequate for proof of design content feasibility and shall allow for finalisation of locations, selections, support and constructability considerations to be incorporated as part of a shop drawing process.

Where implemented, the design team BIM Management Plan has determined the project specific extent of modelling required to achieve project objectives. This may have included documenting aspirational Level of Development (LOD) targets for specific element types. The Consultant Engineers model elements allow for a competitive Tender process (multiple equipment suppliers) and as such have not been nominated as specific elements.

No information contained within the Consultant Engineers Model(s) shall be used by any party for any purpose unless specifically referenced and agreed to in the BIM Management Plan.

The Consultant Engineers Model shall under no circumstances be used for any construction purposes.

In the absence of a design team BIM Management Plan, only the Consultant Engineers 2D Drawings in lieu of the Model shall be used for Tender Documentation or project design communication.

Where the Consultant Engineers Model(s) is provided to the Contractor, it will be done so for information purposes only, in order to assist in gaining general understanding of extent of services elements when combined with the Architectural and Structural federated model. Accordingly it shall be used as a guide together with the Consultant Engineers Drawings and Specifications for the production of shop drawings after:

- a determination of each trades installation methodology
- b safety in design analysis
- c sequence of works considerations
- d final selection of equipment and specific manufacturer recommended installation method plus maintenance, servicing and access needs
- e final arrangement of equipment
- f inter-trades coordination.

Consultant Engineer's coordination activities of services during the design phase was focused on the following:

- g Spatial sizing of major plant to provide spatial allowance only - Contractor shop drawing process and final equipment selection will determine the final equipment set out.
- h Spatial sizing of the services risers - Contractor shop drawing process and final equipment selection will determine the final riser size, installation method and set out.
- i General determination of ceiling zone for allocation of services - Contractor shop drawing process and final equipment selection will determine the final in-ceiling services installation and equipment set out based on site conditions.

Allow in the tender for all costs in related activities to produce coordinated shop drawings for all trades based on the Contract documents, including, but not limited to any requirements listed in a project or construction BIM Management Plan.

3.8 Site Provisions and Building Works in Connection

- 1 This clause is provided to highlight the project requirements in relation to the Site provisions as well as Building Works in Connection BWIC and to ensure that they have been allowed for part of the Contractor Tender submission. The Contractor is responsible for demarcation of tasks and responsibilities between various trades in relation to these items to suit his/her delivery methodology.

Site Provision include but not limited to;

- a Hoisting and Lowering;

- b** Scaffolding:
- c** Temporary Services – Water, Electricity:
- d** Site facilities including Sheds.
- e** Power & Fuel for Testing and Commissioning.

3.8.1 *Building works in connection*

Allow for the following:

1 Demolition:

- a** Services decommissioning, disconnection from supply, capping and sealing.
- b** Services demolition and removal.

Ground Works: Services trenches; excavation and backfilling.

Concrete:

- a** Chasing, coring, cutting and making good.
- b** Placing, casting in and protection of pipe sleeves and conduits.
- c** Bases and plinths for equipment (excluding galvanised steel edge surround to be provided as part of the Hydraulics Services Works Package).
- d** Water-proof curbs round floor penetration in plant rooms.

Masonry:

- a** Chasing, coring, cutting and making good.
- b** Plant enclosures, masonry air ducts and service risers.

Structural Steel:

- a** Lifting beams.
- b** Platforms and walkways for maintenance to equipment.

Roofing:

- a** Openings and under flashings including over flashings
- b** Services works.
- c** Roof access walkways.

Project Requirements and Description of Works

External Walls

- a Setout of cut-outs including cut-out and trimming of openings

Internal walls

- a Setout of cut-outs including cut-out and trimming of openings

Doors:

- a Cut-outs in doors and trimming of openings
- b Undercutting of doors for ventilation purposes.
- c Access doors and hatches for maintenance of Services equipment.

Ceilings:

- a Removal and replacement of ceiling tiles.
- b Co-ordination of penetrations.
- c Ceiling access panels to equipment, valves, dampers and duct access panels above solid ceilings.
- d Sealed air Plenums for smoke exhaust systems.

Dry wall/ Plenum Construction

- a Sealed air plenums for supply and exhaust systems where show on the drawings, including fire rating and thermal insulation where required.
- b Co-ordination of penetrations.
- c Vapour barriers within walls where required.

25mm thick waterproof ply supports for sanitary fixtures on dry wall construction.

Access doors and hatches for access to hydraulic equipment.

Bollards and mechanical protection to pipework, stacks and equipment.

Roof, Gutters, Gutter Sumps, Gutter outlet and external downpipes.

Supply and installation of waterproof membranes in all wet areas.

Set out of building grids to allow set out of core holes.

Supply and install all roof/parapet/balcony overflows.

All in-situ concrete sumps and grated drains (grates and frames to be supplied and installed part of the Hydraulics Works Package) that form part of the structural slab.

Concrete pump plinths. Note: Metal forms to be provided part of the Hydraulics Works Package.

Pipe penetration guards for pipes in carpark

3.9 Services Interface Works

- 1 Contractor is responsible to carry out the works outlined under all contract documents including this Specification to achieve a fully operational installation including all required interfaces between various systems. Whilst the responsibility for interface works and division of works is solely with Contractor the following information is provided for guidance only. Contractor has full control to alter and modify the trade allocation/demarcation of interface works to suit the Contractor's overall programme and delivery methodology in order to achieve a fully integrated, interfaced and operational whole services installations, systems and sub-systems.

Table below indicate key areas where different work packages will interface with each other. Column 1 indicate the disciplines/work packages identified and columns 2 & 3 outline the key interface works between these two work packages.

	DISCIPLINE 1	DISCIPLINE 2
Hyd1/ Mech2	— Provision of HLI or LLI for generating status of equipment on BMS including but not limited to: — Pump Control panels LLI — Meters HLI, — Sensors LLI — Pits — Tanks level sensors — Valves status — Coordinate location and provision of open-end tundishes for all mechanical equipment. — Provision of water supply including back flow prevention to mechanical plant. Terminate with valved branch.	— BMS cabling from Water meters and other BMS monitoring switchboards/control panels/tanks locations back to BMS controllers and specific BMS graphic screens to demonstrate facilities water metering and consumptions in logical subdivisions — Connection of mechanical drains (via insulated condensate / drainage pipework) from mechanical equipment to tundishes. — Extension from valved branches and final connections of water supplies to mechanical equipment. — All required software, hardware, functional descriptions, programming, trending, alarming, testing and commissioning, ITPs, warranty, maintenance, DLP associated with BMS / BAS interfaces and all associated graphic screens to depict such interfaces clearly. — Co-ordinate with the hydraulic contractor and provide drain down flow rate control from mechanical equipment that contain storage water, to limit the output flow rate, so as not to overload the downstream drainage system.

	DISCIPLINE 1	DISCIPLINE 2
Hyd1/ Elec 2	— Coordinate final hydraulic equipment selections and electrical requirements at all provided hydraulic equipment control panels and hard wired connections. — Pressure pumps — Hot water units — Stormwater/sanitary/trade waste pump stations — Hot water circulating pumps — Urinal solenoid mains flush — Electronic tapware — Boiling/chilled water units	— Coordinate and advise after final equipment selections the MD for each hydraulic control panel to ensure correct submain cable selections for the respective switch panel and DB. — Provision of all lugs, cable glands and gland plates for submain terminations and provision of final connections of submains to all control panels and equipment.
Fire1 /Hyd2	— Wiring and connection to the FIP of monitoring devices on nominated isolating valves associated with fire protection system water supplies.	— Provision of isolating valves complete with electronic monitoring devices (Class A) on incoming fire hydrant and sprinkler water supplies. — Provision of gas shut off solenoid
Hyd1 /VT2	— Provision of drainage sumps Inspection opening to surface in lift pits as required. Manual pump out as required	

4 Design and Performance Criteria

4.1 General

- 1 The following standards are applicable throughout the project.
 - a National Construction Code 2022 (Referred throughout this specification as NCC) including all relevant Australian Standards called in the NCC
 - b National Construction Code (NCC) Part 3 – Plumbing Code of Australia (PCA)
 - c Relevant Occupation/Workplace Health and Safety legislation
 - d The Environmental Protection Authority
 - e WorkCover
 - f Any other Authority having jurisdiction over all or part of the installation to ensure that the equipment and installation, when manufactured and installed, will comply with the rules and regulations.
 - g Queensland Water Act.
 - h Queensland Plumbing and Drainage Act.
 - i Queensland Development Code.

The Hydraulic Services shall be to the approval of:

- j Rockhampton Regional Council.
- k Fitzroy River Water.
- l Queensland Fire Department.

The following standards are applicable throughout the project as a minimum. Where items of equipment are required the relevant Australian Standard to the equipment will apply. If clarification required, seek clarification at the time of tender.

Where Australian Standards and Codes do not exist, the appropriate British Standard or Code shall apply.

4.2 Seismic Restraints

4.2.1 Design standard

- 1 NCC – Building Code of Australia.

AS1170.4 – Structural design actions – Earthquake actions in Australia.

4.2.2 Design parameters

- 1 The following design parameters shall be used to design the Engineering Services equipment and fastenings as per AS1170.4:

SEISMIC DESIGN PARAMETERS	
Importance Level (IL)	Refer to Structural Engineering Documentation Package
Site Sub-Soil Class	Refer to Structural Engineering Documentation Package
Hazard Factor (Z)	Refer to Structural Engineering Documentation Package
Probability Factor (Kp)	Refer to Structural Engineering Documentation Package
Earthquake Design Category (EDC)	Refer to Structural Engineering Documentation Package

- 2 Design and install the scope of works to withstand the earthquake forces determined from the above design criteria in accordance with the requirements of the BCA and AS1170.4.

4.3 Design Criteria

ITEM	AUSTRALIAN STANDARDS	DESIGN CRITERIA	
Water Supply			
	AS/NZS 3500.1 Water Services	To local Water Authority Standards In accordance with AS 6400 – Water Efficient Rating and Labelling Plumbing Code of Australia	
Supply Conditions		Max Water design velocity within Building	2.1 m/s
		Max pressure at any outlet	500 kPa
		Min pressure at any outlet	250 kPa
Pipe sizing design criteria - Hot & Cold	Australian Standard 2200	The Institute of Plumbing Australia <i>(Selection and Sizing of Copper Tube for Water Piping Systems. B Smith)</i>	
Outlets	AS1428.1 Design for Access and mobility	WaterMark TM	
Water conservation measures	AS 6400	Water Efficiency Labelling and Standards (WELS) Scheme	
Heating Hot Water			
	AS/NZS 3500.4 Heated Water Service	To Local Authority Standards	
Supply Conditions		Stored HW temperature	65°C
		Max personal hygiene delivery temp	50°C
		Max personal hygiene delivery temp for aged, sick, children or people with disabilities	42°C
		Max allowable flow velocity (circulatory)	1.2 m/s
		Max allowable return velocity (circulatory)	1.0 m/s
		Max allowable flow velocity (other)	2.1 m/s
		Max temp loss through circulatory system	5°C
		Max dead leg length (total volume)	2 Litres
Heated water temperature control	AS/NZS 3500.4 Heated Water Service	To local Authority Standards	
Hot Water Plant	AS/NZS 3500.4 Heated Water Service	Hot Water Plant to be sized to meet peak hourly demand and daily demand N+1 redundancy shall be provided within each plant of 1 additional component (1 x burner and 1 x storage tank)	

ITEM	AUSTRALIAN STANDARDS	DESIGN CRITERIA	
Sanitary			
	AS/NZS 3500.2 Sanitary Plumbing and Drainage	To local Authority Standards	
General Discharge Conditions		All sanitary drainage to grey water treatment plant and/or discharge off site via authority sewer connections.	
Drainage Systems		The drainage system shall be designed to allow the removal of foul gases to atmosphere via an open vent pipe A barrier or water seal shall be maintained at each fixture to prevent foul gases entering the building environment.	
Sanitary pump Stations		Type	Submersible pumps
		Tank level monitoring	Via sensors calibrated for effluent level (%)
		Tank level warnings	High level
		Emergency storage	Min 1 hour at peak inflow
		Pump duty	Duty/standby
		Pump Level Control	— Duty pump stop — Duty pump start — Standby pump start — High level alarm
		Redundancy	N+1 redundancy shall be provided
Trade waste			
	AS/NZS 3500.2 Sanitary Plumbing and Drainage	Industrial and Trade waste drainage to the Local Authority requirements. Sewer Waste Characteristics including — Hydrocarbon based product — Fats Oils and Grease (FOG) — BOD — TSS — pH shall be treated or removed to approved Local Authority limits	
Treatment	Queensland Urban Utilities in lieu of local authority sizing requirements – Trade waste sizing grease and oil arrestors	Type	Oil/Water Separator and Grease Arrestor
		Treatment capacity	Meet final hourly tradewaste volumes established in accordance with local authority tradewaste guidelines
		Buffer Tank effective volume	100% of peak hourly treatment capacity

ITEM	AUSTRALIAN STANDARDS	DESIGN CRITERIA	
		Tank level monitoring	Via sensors calibrated for effluent level (%)
		Tank level warnings	Mid level, high level
		Emergency storage	Min 1 hour at peak inflow
		Monitoring of end quality	Automated
Gas			
	AS 5601.1 Gas Installations	Mains supply and delivery to meet the Gas Authority and network operator. Gas meters and pressure reduction stations to meet the Gas authority and network provider	
Stormwater			
	AS/NZS 3500.3 Stormwater Plumbing and Drainage	To local Authority Standards The stormwater installation to cater for the Annual Rainfall Intensity (ARI) for the site	
Discharge Conditions		To AS/NZS 3500.3 and local authority requirements	
Stormwater pump stations		Type	Surface pumps
		Tank level monitoring	Via sensors calibrated for effluent level (%)
		Emergency storage	As per schedules
		Pump duty	Duty/standby/assist
		Pump Level Control	— Duty pump stop — Duty pump start — Standby pump start
		Redundancy	N+1 redundancy shall be provided

4.4 Design Assumptions/Parameters

In the absence of adequate site survey data and recognising that available as-constructed drawings may be outdated or in some cases not available, this section outlines the key assumptions required to progress the design. These assumptions are based on engineering judgement and the best information currently accessible and will be refined as updated records or detailed site investigations become available. The following assumptions have been adopted to facilitate ongoing project development.

The key assumptions made:

- In-ground services can traverse unimpeded in the paths documented
- Site sanitary/stormwater drainage system has adequate capacity and compliant gradient and cover
- Site sanitary/stormwater drainage pipework is sound and fit for purpose
- Sanitary/Stormwater drainage services adjacent to buildings have invert levels suitable for building drainage connection via gravity.
- Site sanitary drainage is privately owned/maintained.
- Only new buildings will require rainwater harvesting and reuse in accordance with the ESD report.
- 5 microns of filtration will supply acceptable sediment filtration for rainwater reuse.
- Where pump stations are being utilised, they will discharge to nearest live gravity drain.
- Water grade being supplied to the site is of an acceptable grade for end user.
- Water services serving the areas of work have available minimum capacity of 500kPa at 2.0 L/s to the area of work.
- Water services serving the areas of work have adequate cover, where located below ground.
- Water services will be provided with isolation valves to each area.
- Centralised hot water supply only where directed by the client
- Hot water will have 1 hour of peak demand twice a day, where a centralised plant is used.
 - o Building 33 - 5,000 L for the first hour and recovers at 190 L/hour
 - o Building 36 – 1,125 L for the first hour and recovers at 298 L/hour
- Gas emergency shut-off upon triggering of fire alarms is not required.
- All gas equipment will be installed with flame sensors or thermocouples

4.5 Environmental Rating Systems Requirements

The following additional requirements are to be met if required by the Green Star Schedule in Schedules.

- 1 Comprehensive pre-commissioning, commissioning and quality monitoring are to be performed in accordance with ASHRAE Guideline 1 or CIBSE Commissioning Codes.

Provide a complete commissioning report which confirms that all commissioning is in compliance with the contract documents. The report must also include a list of outstanding commissioning issues, records of all functional/commissioning testing undertaken and list any seasonal testing required in the future.

Owner instruction to be spread over at least two sessions separated by at least two months. Issues covered must include:

- a Energy and environmental features of the design.
- b Review of the controls set-up, programming, alarms and trouble shooting.
- c Operating and Maintenance manuals.
- d Building operation including start-up, normal operation, unoccupied operation, seasonal changeover and shutdown.
- e Interactions with other systems.
- f Optimising energy efficiency.
- g OH&S issues.
- h Maintenance requirements and sourcing replacements.
- i Occupant satisfaction feedback.

Allow to coordinate the above with electrical and hydraulics Works Packages so that a whole-building approach is achieved.

Separately from the routine maintenance and servicing, provide detailed formal monthly reviews of the Hydraulic Services provide written reports to the building owner quarterly detailing the status of the system, changes made, etc. As a minimum, each review shall include:

- a All system set points and timers.
- b Rectification of all BMS alarms and other faults.
- c Full load commissioning tests.
- d Minor adjustments to BMS programmes, set points, water flows, etc. as found necessary from operating experience or as advised by the Consultant Engineer to time the system for optimum performance and energy efficiency.

Provide a formal final recommissioning of the hydraulic services in the final month before Final Completion (12 months after Practical Completion). As a minimum, this is to include:

- a A written report to the building owner as per the quarterly review.
- b All quarterly issues.

Engagement of an Independent Commissioning Agent (ICA) as per Green Star requirements whose role will be to provide commissioning advice to the design team and to the Building Owner and to monitor and verify the commissioning of the HVAC and building control systems. The independent commissioning agent will report directly to the Building Owner and the primary goal of the agent is to independently identify system deficiencies as early in the project as possible and track their status until corrected. The ICA's role and responsibilities include:

Be an objective advocate of the Building Owner.

- a Review basis of design and design intent as well as preliminary working drawings.
- b Create a commissioning plan and direct the commissioning process that covers the following (if present): grey water treatment plant, hot water plant, thermostatic mixing valves, automatic controls, hot and cold water systems, air admittance valves.
- c Introduce standards and strategies early in the design process.
- d Set target requirements in the contract documents to ensure implementation of selected measures.
- e Coordinate with the Building Owner, design team and Contractor throughout the commissioning, testing and adjustment phases.
- f Observe all testing, review and approve Contractor test results.

Prepare final commissioning report including recommendations to the Building Owner regarding the performance of commissioned building system.

4.5.1 *Relevant green star compliance requirements*

- 1 Reference Green Star Office Design V2.

Design and Performance Criteria

Details scheduled hereunder form the basis of design and are required to be maintained.

Refer to Green Star Manual and to Specification sections for additional details.

Overall Project Required Green Star Rating TBC

CREDIT NO.	POINTS	NOTES
Man-2	TBC	Refer to ESD Consultants Documentation Package
Man-3	TBC	Refer to ESD Consultants Documentation Package
Man-4	TBC	Refer to ESD Consultants Documentation Package
Man-7	TBC	Refer to ESD Consultants Documentation Package
IEQ-1	TBC	Refer to ESD Consultants Documentation Package
IEQ-2	TBC	Refer to ESD Consultants Documentation Package
IEQ-3	TBC	Refer to ESD Consultants Documentation Package
IEQ-10	TBC	Refer to ESD Consultants Documentation Package
IEQ-12	TBC	Refer to ESD Consultants Documentation Package
IEQ-13	TBC	Refer to ESD Consultants Documentation Package
IEQ-15	TBC	Refer to ESD Consultants Documentation Package
IEQ-16	TBC	Refer to ESD Consultants Documentation Package
Ene-1	TBC	Refer to ESD Consultants Documentation Package
Ene-2	TBC	Refer to ESD Consultants Documentation Package
Ene-3	TBC	Refer to ESD Consultants Documentation Package
Ene-4	TBC	Refer to ESD Consultants Documentation Package
Wat-2	TBC	Refer to ESD Consultants Documentation Package
Wat-4	TBC	Refer to ESD Consultants Documentation Package
Mat-7	TBC	Refer to ESD Consultants Documentation Package
Mat-8	TBC	Refer to ESD Consultants Documentation Package
Emi-1	TBC	Refer to ESD Consultants Documentation Package
Emi-2	TBC	Refer to ESD Consultants Documentation Package
Emi-3	TBC	Refer to ESD Consultants Documentation Package
Emi-4	TBC	Refer to ESD Consultants Documentation Package
Emi-9	TBC	Refer to ESD Consultants Documentation Package
Inn-1	TBC	Refer to ESD Consultants Documentation Package
Inn-2	TBC	Refer to ESD Consultants Documentation Package
Inn-3	TBC	Refer to ESD Consultants Documentation Package

5 Painting, Protection and Identification

5.1 General Requirements

1 Select painting and finishes in accordance with the area environmental conditions.

Review the risk assessment and statutory and authority requirements completed as part of the system testing, commissioning and handover requirements and provide labelling and signage as determined.

Where exposed to view paint exposed equipment, cable trays, ductwork and pipework.

Where ductwork, pipework, cable trays exposed to view is not specified to be painted, all factory and installation markings shall be removed.

All black steel surfaces and surfaces subject to corrosion shall be primed with at least two coats of primer and protected.

Protect all items of work during dispatch and whilst on site during progress of installation.

Fabricated pipe and steelwork shall be delivered to site prime coated..

Concealed pipe work to be prime coated

If exposed to view, paint new services and equipment including in plant rooms, except chromium, anodised aluminium, GRP, UPVC, stainless steel, non-metallic flexible materials and normally lubricated machined surfaces. Repaint proprietary items only if damaged.

Do not combine paints from different manufacturers in a paint system.

Remove or protect fixtures, equipment, surfaces and labels before starting to paint, and re-instate after completion of painting.

All internal surfaces of air outlets and openings exposed to view shall be painted matt black.

Repair factory finishes if damaged with identical finishes.

Finish visible joints made by welding, brazing or soldering using methods appropriate to the class of work (including grinding or buffing) before further treatment such as filling, painting, galvanising or electroplating.

If galvanised surfaces have been cut or welded after galvanising, prime the affected area using zinc rich organic binder AS/NZS 2312.

Before applying coatings to metalwork, complete cutting, drilling and other fabrication, remove all grease, oil and other contaminants and prepare surfaces to AS 1627.0.

Use relevant Australian Standards for the following finishes; Galvanising, Electroplating, Anodising, Thermoset powder coating, Two-pack liquid coating, Air-drying enamel, Stoving enamel.

5.2 Submissions

1 Provide the following submissions or samples:

- a Proposed Labelling Systems.
- b Schedule of statutory, maintenance and operating signage.
- c Valve Identification.

5.3 Corrosion Protection

5.3.1 General

1 Apply corrosion protection measures to all metallic surfaces.

Select an appropriate protective coating system to suit atmospheric corrosivity category as defined in referenced Standards/Codes.

Fastenings are to be protected to have a corrosion resistance equivalent to the parts fastened.

All welding is to be completed before any corrosion protection treatment is applied.

5.4 Painting

5.4.1 Low VOC

The following additional requirements are to be met if required by the Green Star Schedule in Schedules.

1 All painted surfaces to use low-VOC paints.

VOCs are to be in accordance with The Australian Environmental Labelling Association, Inc. Standard No: AELA 23-2005 'Australian Voluntary Environmental Labelling Standard Architectural and Protective Coatings'. Conformance with VOC levels listed in this standard (refer to table below) shall be demonstrated by providing test reports from laboratories accredited to carry out the relevant tests and/or calculations and appropriate documentation of production methods and quality controls.

PRODUCT TYPE	MAXIMUM VOC CONTENT (G/LITRE)
Latex Primer for galvanised iron and zincalume	60
Exterior latex undercoat	55
Interior latex undercoat	65
Interior gloss	75
Interior semi-gloss	16
Exterior gloss	75

VOC limits on architectural coatings covered. Allowable levels include water content in the formulation.

For solvent-based coatings the paint shall not contain VOCs in excess of 200g/litre. For recycled paints the VOC level (averaged across batches) in the paint must not exceed 100g/litre.

The VOC content of the paint shall either be calculated from the VOC data for each of the raw materials or, experimentally by ASTM D3960, as qualified the Australian Environmental Labelling Association, Inc. Standard No: AELA 23-2005 'Australian Voluntary Environmental Labelling Standard Architectural and Protective Coatings'.

Where the raw material is a mixture of compounds, some of which contain VOCs, the VOC content of the mixture may in turn be calculated from the VOC content of the individual components. Where this is not known, it must be determined by the methodology detailed in AELA 23-2005.

5.4.2 Preparation of surfaces

1 Preparation: all surfaces shall be prepared by thorough cleaning of all dirt, grease, scale, welding flux and corrosion in accordance with AS/NZS 2312.

For galvanised surfaces that have been subsequently cut or welded, prime the affected area with organic zinc rich coating for protection of steel to AS/NZS2312, two pack or accepted alternative.

Prime/undercoat all surfaces. Galvanised and non-ferrous surfaces must be etch primed.

5.4.3 *Application*

- 1 First coat: apply the first coat immediately after the substrate preparation and before contamination of the substrate can occur.

Apply all paint systems in strict accordance with manufacturer's recommendations.

Finish: each coat of paint or finish shall be uniform in colour, gloss, thickness and texture and free of runs, sags or blisters.

5.4.4 *Finishing*

- 1 Concrete Bases: 2 coats synthetic emulsion paint.

Equipment Panels/Switchboards: 2 final coats full gloss enamel.

Equipment: Factory spray application.

Other surfaces: 2 final coats full gloss enamel.

5.5 Colours

Paint colours nominated are to be in accordance with AS 2700 and AS 1345.

5.6 Identification and Labelling

5.6.1 *General*

- 1 Mark all systems and equipment including wiring, piping, control panels, switchboards, ductwork, equipment cabinets, conduits, controls, gauges, valves and fittings to that they are readily identifiable.

Equipment requiring special maintenance procedures or presenting occupational health and safety hazards: provide permanent labels complying with statutory requirements.

Locate labels so that they are easily seen and either attached to, below or next to item referred.

For systems containing hazardous materials provide labelling in accordance with AS 1318.

Where a pipe or duct passes through a wall, slab or partition provide labelling each side.

Equipment and Service labels shall be in accordance with the Legend of Symbols nomenclature for the respective service. Provide appropriate Emergency and Operating Instructions to every equipment and system.

Provide text and simple diagrams where practicable. Locate at a fixed point and point of operation.

Affix hazardous equipment with warning signs.

Labels shall be permanent.

Type: select from the following:

- a Engraved and black filled lettering on stainless steel or brass plate, minimum 1mm thick mechanically fixed.
- b Stencil with black or white lettering contrasting with background.
- c Engraved two-colour laminated plastic mechanically fixed.
- d Adhesive labelling.
- e Painted.

Adhesive labels and laminated plastic labels shall not be used where exposed to the environment.

Painted labels shall be in the same finish paint type as the installation.

5.6.2 Labelling requirements

ITEMS	REQUIREMENTS
Pipework (Hyd)	Labelling and lettering to AS 1345 and AS1318
Conduits	Labelling and lettering to AS 1345
Exposed Conduits	Labelling and lettering to AS 1345
Valves	Labels shall be colour coded laminated labels indicating service, function and normal position, and attached to hand wheels or spindles using a vandal resistant permanent chain/ring
Switchboards(general)	— Main switches: 10mm. — Outgoing electrical functional units: 8mm. — Automatic controls electrical equipment and instruments: 5mm. — Rear covers of switchboards: 5mm. — Components inside electrical enclosures and control panels: 3.5mm — Locate labels so that they are easily seen from normal access adjacent to the item being marked. Do not install labels on components normally removed or replaced.
All equipment such as pumps, tanks, hot water plant including controls, gauges and fittings to that they are readily identifiable	— Label in accordance with the specification abbreviations, symbols and acronyms scheduled or otherwise approved. — Locate labels so that they are easily seen from normal access adjacent to the item being marked. Do not install labels on components normally removed or replaced. — Major equipment nameplates: 40mm. — Minor equipment nameplates: 20mm. — Danger, warning and caution notices: 10mm for heading, 5mm for text. — Warning notices: 7mm. — Minor lettering: 3mm. — Lettering Style: Helvetica Medium. — Fixing: Use mechanical fixings. Do not penetrate isolation vapour barriers. — Equipment requiring special maintenance procedures or presenting occupational health and safety hazards: provide permanent labels complying with statutory requirements. — Equipment requiring consumables including replacement belts, oils, filters and strainers: provide labels indicating consumable component details and quantities and corrective maintenance trigger point such as filter pressure drop.

6 Inspection, Testing, Commissioning, Handover and Defects Liability Period

6.1 General

- 1 This section sets out the engineering and quality assurance requirements for the completion of the installation, inspection regime, testing, commissioning, handover and defects liability period.

6.1.1 *Design*

- 1 Complete the detail engineering of the testing commissioning and handover of the works to the specified requirements.

Comply with applicable authority design guidelines and codes of practice and requirements.

As a guide, commissioning guidelines as detailed by CIBSE Commissioning Codes shall be used as a minimum to undertake the commissioning of services listed in this section.

6.1.2 *Standards & guidelines*

- 1 Refer to Design and Performance Criteria section

6.1.3 *Practical completion requirements*

- 1 Notwithstanding any other provisions of the Contract with regard to the granting of Practical Completion, the following requirements, listed in order of importance / priority, shall be completed and completion submissions made prior the date of Practical Completion:

Minimum submissions required for application for Occupancy Permit.

- 1 Submission of signed off testing and commissioning schedules to confirm that Testing and Commissioning has been finalised and that all plants and systems operate in a safe, stable and automatic manner under all conditions of full and partial load - full commissioning data.

Submission of Authority approvals including Statutory Authority approvals and Certification of Compliance with the NCC.

Submission of Certification that works comply with the Contract Documents.

Operating Instruction and Maintenance manual sections covering all Safety Services as required by regulations.

Other submissions:

Submission of correctly and completely executed ITPs including test results completed in accordance with the Contractor Quality Plan and the satisfaction of the Consultant Engineer.

Adequate training and instruction of the Principal's representatives in safe operation of the plant.

Official Equipment manufacturer's acceptance certification for the installation.

Completed Operating and Maintenance manuals and 'as built' drawings

The Defects Liability period may be extended in the event:

- a Specific systems fail to achieve the desired outcomes after two attempts are made to rectify the drawback. These specific systems include those that are fire and life safety or business / mission critical to the Principal.

- b** If there have been more than three (3) attempts to rectify the defect, in which instance a further 12 months of monitoring will be required after the defect is rectified to monitor that the desired outcomes are achieved without further shortfalls.

6.2 Practical Completion Requirements

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6.3 Management of Inspection, Testing, Commissioning, Handover and Defects Liability Period Commitments

6.3.1 General

- 1** Manage the whole process:
 - a** Ensure complete execution of Project Specific Quality Plan and specifically Installation and Commissioning ITPs are completed progressively from commencement of the project.
 - b** Test and verify the operation of individual items of plant and equipment, sub-systems, systems and the overall installation including the interfaces with other building services systems forming part of the project.
 - c** Create and record accurately and verify all "As Built" drawings progressively prior to services being concealed by construction of ceilings, wall cavities and shafts.
 - d** Ensure completed Operating and Maintenance manuals inclusive of comprehensive system descriptions, operating modes and associated control functionality, tailored and project specific manufacturers literature.

- e Responsive defects rectification process during the construction phase as well as post Practical Completion during Defects Liability Period.
- f Ensure Responsive approach towards maintenance and operational fine tuning during the Defects Liability Period

6.3.2 Sequence of activities

- 1 Generally, follow the sequence of activities listed below. Modify to suit specific requirements of the project however no reduction of scope is allowed.
 - a Submission of Project specific Project Quality Plan including generic and project Specific ITPs within the first quarter of the project programme after site possession.
 - b Ongoing inspection of installation works by relevant Quality Manager and/or his nominated staff for meeting project quality requirements including preparing Inspection reports (listing all observed installation defects) not longer than fortnightly and also checking that all Installation ITPs are signed off.
 - c Initiate Safety in Design process and documentation where applicable
 - d Evidence of adequate commissioning programme allowance in construction programme.
 - e Production and submission of commissioning Inspection and Test Plan (ITP) 12 weeks after site possession
 - f Amendment to ITP as necessary and re-submission at least 1-2 weeks after the original submission.
 - g Submission of controls manufacturer's functional descriptions elaborating on the methodology proposed to implement the requirements of the control strategies the manufacturer's controls functional descriptions shall be prepared and submitted at least 12 weeks prior to start of installation and commissioning of controls systems.
 - h Nomination of those to be involved in commissioning including individuals with responsibility for management, engineering and field testing and commissioning.
 - i Liaison with and briefing of Authorities to ensure that the connection to their assets and commissioning procedures and resultant data provided for their approval is consistent with their requirements and records.
 - j Preparation of periodic (minimum monthly) progress reports on testing and commissioning.
 - k Early and active engagement of Independent Commissioning Agent ICA where applicable.
 - l Early (within the first quarter of the project construction programme) submissions of Operating and Maintenance Manual (for Safety Services and, for preliminary draft of the overall manual). Confirmation of acceptance of proposed Electronic Operating and Maintenance Manual software/platform.
 - m Submission of detailed commissioning procedures for review by the Consultant Engineer. Submission shall be made early, at least 12 weeks prior to the commencement of commissioning when the project construction period greater than 12 months, or 6 weeks when the project construction period is less than 12 months' duration.

Rectification of all defects likely to interfere with testing and commissioning.

Commencement of "As Built" drawing production.

Carryout preliminary commissioning including pre-commissioning procedures until achievement of correct operation and performance. Providing statement that the pre-commissioning activities for each system are completed.

Testing and commissioning of major plant and equipment, complete systems, interfaces between systems and finally all systems which interact together under normal or emergency conditions.

Integrated testing and commissioning of multi systems (including multi-disciplinary systems) and their associated interfaces.

Inclusion of all completed Commissioning ITPs in the Operating and Maintenance Manual.

Rectification of any further defects and deficiencies found during testing and commissioning.

Provide notification of completion of remedial work in sufficient time to permit re-inspection as necessary before the intended date for re-testing.

Repetition of sequence of activities as necessary until works comply with acceptance criteria as set out in the ITPs.

Final handover of Final Operating and Maintenance Manuals.

Certification that works comply with the Contract Documents and with the NCC and Statutory Authorities and are ready for granting of Practical Completion.

6.3.3 Submissions

- 1 Required submissions include but not limited to the following:
 - a Installation and commissioning Inspection and Test Plans ITPs. 4 Copies of ITPs to be supplied, 3 bound into Installation Manuals and 1 bound separately.
 - b Evidence of periodic (weekly) Quality Inspection/defects reports.
 - c Testing and commissioning procedures including integrated commissioning procedures.
 - d Evidence of Safety in Design process during the construction phase is completed where applicable.
 - e Evidence of currency of calibration of equipment to be used in commissioning procedures.
 - f Records of all pre-commissioning checks and final commissioning data. All records/data to be certified as being conducted in accordance with agreed and approved procedures.
 - g Manufacturers' certificates for all proprietary items, to confirm that the installation complies with their installation and maintenance requirements prior to start-up of equipment. Certificates shall also be provided at commencement of equipment installation to confirm their requirements with respect to installation are being provided for.
 - h Manufacturers' test certificates of all capacity/performance verification checks carried out at factory prior to dispatch to site.
 - i Accurate and verified (by QA manager) "As Built" drawings.
 - j Complete Operating and Maintenance Manuals (Electronic system/package and, unless instructed otherwise, hard copies).
 - k Warranties.
 - l Maintenance Plan and Schedules following acceptance of technical submission and installation minimum 4-8 weeks prior to Practical Completion with priority with respect to timely completion given to plan and schedules covering Safety Services.

6.3.4 Inspection and test plans

- 1 Inspection and test plans are to be prepared specifically for the project but may be based on or customised from generic ITPs. All commissioning result forms shall be completed, including the following:

Tested by: please print	Witnessed by:
Signed:	Signed:
Date:	Date:
Test Device used:	
Serial number:	
Date of Calibration:	
Method of Testing:	

List acceptance criteria for each element and sub-element of the installation in ITPs. Acceptance criteria to include:

- a Specification details in regard to materials, construction methods, physical requirements, performance and operational requirements.
- b Schedule of Technical Data in respect of make, model and performance details of equipment.
- c Shop drawings.
- d Control logic and diagrams.
- e Authority inspection and testing requirements.

- f References to specific clauses of this Specification for acceptance criteria

6.3.5 *Commissioning instrumentation*

- 1 All instrumentation used in the commissioning of the installation shall be managed in accordance with the requirements of AS 3912 or NEBB.

Reference instruments are to be NATA certified.

All commissioning results to include instrument calibration documentation.

6.3.6 *Commissioning personnel*

- 1 Provide only qualified registered / licensed personnel specific to the trades as required by local regulations, and where regulatory requirements do not apply, personnel certified by appropriate industry bodies.

Submit a schedule of key commissioning personnel together with their qualifications prior to undertaking commissioning activities on site.

6.4 Testing and Commissioning

6.4.1 *Pre-energisation tests*

- 1 Prior to energisation of any system with electrical connections, conduct the following tests:
 - a Tightness of screwed and bolted connections.
 - b Physical integrity.
 - c Motor rotation.
 - d Insulation resistance tests.
 - e Test operation of all trips, interlocks, motor driven devices, contactors and control circuits and devices by instigating or simulating inputs.

6.4.2 *Tests*

- 1 Include the following in testing of installation:
 - a Pump installation including all pump controls and interfaces.
 - b Domestic hot water plant.
 - c Grey water Treatment plant and all interfaces
 - d Backflow prevention devices
 - e Ultraviolet disinfection plant
 - f Filtrations systems.
 - g BMS interface
 - h Hot water circulatory and solar transfer pumps.
 - i Interface to the BMS
 - j Thermostatic valves
 - k Tapware operation.
 - l Downpipe/overflow, planter drains, waste and drain systems clear of blockage.
 - m Syphonic stormwater outlets clear of blockage.
 - n Plastic inspection opening covers have been replaced with final chrome plated brass covers.
 - o Sanitary and Stormwater pump station chamber and pump level indicators, check all BMS interfaces and offline power interface
 - p Gas meters and submeters, check gas meter emergency safety shut off and isolation working correctly
 - q Sanitary plumbing-Provide a hydrostatic test to maximum choke level to the satisfaction of the authority
 - r Provide a hydrostatic test to choke level for a minimum period of 15 minutes and as required by the testing authority and the Contractor and as specified under "Preliminaries" section.

- s On completion, all work shall be hydrostatically tested under expected maximum choke condition including box gutter sump and box gutter for a period of two (2) hours
- t Provide a water pressure test of 1500 kPa for a period of two hours. Disconnect any equipment connected to the service not rated to the test pressure before testing commences.”
- u Natural Gas-Test all pipework 1700 kPa for a period of two hours. On completion, test the installation under full supply conditions.
- v Provide a water pressure test of 1500 kPa for a period of two hours. Disconnect any equipment connected to the service not rated to the test pressure before testing commences
- w Hydrostatic Tests:
 - i All water piping shall be hydrostatically tested to a head of one and a half times the working head. Hydraulic (testing) pumps shall be disconnected immediately after pressurisation and all test heads shall be maintained until the Contractor has satisfied himself as to the soundness of the pipe work and equipment.
 - ii In no case, shall the period of test be less than 30 minutes.
 - iii Before applying specified test head, all air shall be expelled from the piping being tested. If necessary, supply and install approved air relief valves for this purpose.
 - iv Equipment is not to be connected to the respective services while hydrostatic tests are being carried out.
 - v Provide a suitable pump and gauge and do all necessary work for carrying out the tests.
 - vi All testing of water piping is to be carried out before:
 - A Finishing trades have commenced their work;
 - B Ceilings are installed;
 - C Insulation of pipe work.
 - vii At completion of testing, all water lines shall be kept charged full of water at all times.
- x Choke Tests:
 - i All sanitary drains, stormwater drains, sanitary plumbing, rainwater pipes, shall be choke (water) tested by plugging at the lowest, or other approved positions and filling with water to the overflow level.
 - ii Choke tests shall be maintained until the Contractor and/or Authorities have satisfied themselves as to the soundness of the pipe work.
 - iii In no case, shall be period of test be less than one hour.
 - iv Fixtures are not to be connected to the respective services while choke tests are being carried out.
 - v Provide all necessary plugs and do all necessary work for the carrying out of the tests.
 - vi All testing of the above system shall be carried out before:
 - A Finishing trades have commenced their work,
 - B Ceilings are installed,
 - C Backfilling has commenced,
 - D Absorption into pipe wall is complete.

y Fixtures:

- i Fixtures shall be filled to spill level with water after installation and visually checked for leaks.

z Test of Completion:

- i Upon completion, the Hydraulic Works Package shall be tested under normal working conditions and as directed by the Contractor; such tests shall continue until the Contractor is satisfied that the terms of this specification has been complied with and that the Hydraulic Works Package are capable of meeting all requirements.
- ii All defects disclosed during the tests shall be remedied immediately and, if required by the Contractor fresh tests shall be carried out.
- iii The duration of the tests will be decided by the Contractor. The maximum duration of any one test will not exceed eight hours.

aa Essential Services:

- i Upon completion of the works, provide Essential Services Certification to the requirements of the National Construction Code for the following:
 - A Fire mains and pumps.
 - B Fire hydrants.
 - C Fire hose reels.
 - D Fire extinguishers (where applicable).
- ii A certificate of compliance for each essential service installed under their contract as per requirements of the NCC is to be obtained and submitted upon completion of the works.

6.4.3 *Pre-commissioning tests*

1 Prior to commencement of the commissioning of the systems, carry out the following:

- a Progressively during the construction phase regularly purge cold water from all systems to ensure that stagnation shall not occur.
- b Clean and purge all cisterns, grease arrestors and pressure vessels, remove all scum lines and high water marks
- c Thoroughly clean out and flush out all water systems to ensure only clean water is within the systems.
- d Pressure and leak test all water systems as outlined in Section 'Pipework, Valves & Fittings'.
- e Thoroughly clean and flush out all water systems including package pressure system.
- f Protect the filtered and treated water systems by the complete flushing out of the system prior to commissioning and installation of the filters.
- g Provide manufacturers test certificates for all proprietary manufactured items including pumps, water filters, backflow preventers, thermostatic valves and Ultraviolet disinfection systems -. Mount test certificates for pressure vessels in glazed frames adjacent the equipment. Fix to pressure relief valves the manufacturers tags indicating the set pressure of the valves and stamp all pressure valves with the manufacturer's design and test pressure ratings.
- h Overhaul all installed backflow preventers including full commissioning of same by manufacturers representative
- i Check and prove all pressure and flow actuated controls, relief devices, bleeds, vents and drains.
- j Fully open all control valves during balancing of water flows. Permanently mark final settings of all regulating valves in an appropriate manner.
- k Check installed thermostatic mixing valves, back flow preventers and trap primers and set by the manufacturer's representative.
- l Check all Backflow prevention devices and record and submit to the appropriate authorities in accordance with manufactures and local authorities' requirements.
- m Check level indicators on storage tanks are monitored correctly by the BMS. High and low level alarms to be generated.

- n Adjust period controls and activate by competent personnel provided by the manufacturer or his authorised agent. Maintain a complete log of the settings and readings of all controls, gauges and instruments throughout the Commissioning period.
- o Check all water meters to be commissioned and monitored via the BMS. Maintain a complete log of the settings and readings throughout the Commissioning period.
- p Adjust level sensors and commission pumps as required. Pumps controlled by controller provided as part of the packaged pump station. BMS to monitor high and low level alarms.
- q Commission all pumps.
- r Check BMS to monitor pump failures (all pumps).
- s Check hot water system and ensure plant is operational and working within to the required peak hourly flow rate. BMS to monitor water meter located on cold water inlet of hot water plant.
- t Grey water plant and all associated interfaces to be commissioned by specialist grey water treatment plant manufacturer and operator. Data from grey water plant control panel to be monitored by BMS.
- u Check the operation of all indicating and recording instruments and associated equipment and the indicating lights and associated flashers and relays under full working conditions.

6.4.4 Commissioning

- 1 Operate the installation to prove the performance, capacities and ability to provide the required service.

Undertake commissioning in 4 stages:

- a Individual component functionality testing including verification of defects free installation
- b Major items of plant and equipment and components.
- c Systems and sub-systems.
- d Inter-system operation and interfaces.

Commissioning to include:

- e Performance of individual plant items and components.
- f Operating sequences, interlocks and safeties.
- g Final controls calibration.
- h System operation under all operating modes and under all conditions of load.
- i Inter-system operation and correct interfacing connections under all operating conditions and under simulated fire conditions.
- j Noise and vibration tests.
- k Thermoscan survey of switchboards.
- l Environmental audit as required by Authorities.
- m Rectification and correction of any defects and deficiencies.
- n Continue commissioning until achievement of correct performance and operation.
- o Electrical testing and verification to AS/NZS 3000.
- p Cleaning of plant room, switchboards and all equipment and devices.

6.5 Final Contract and Handover Documentation

6.5.1 General requirements

- 1 Documentation is required in electronic and hard copy format. Electronic documentation shall be supplied on the latest Microsoft and AutoCAD (or Revit if specified) Software versions at the time of Practical Completion.

Scope of the documentation and format requirements comprise:

DOCUMENT	FORMAT	SETS
	Electronic installed on BAS File Server	1

DOCUMENT	FORMAT	SETS
Operating and Maintenance Manuals, including Final commissioning data	CD ROM – Microsoft Office Latest Version	3
	Hardcopy	3
As Built Drawings	CD ROM – AutoCAD	3
	Hardcopy – Full Size with hanging strips	1
	A3 Size	2
Control Software	CD ROM	1
OH&S Safety Signage	Permanent	1

6.5.2 *As built drawings*

- 1 All As Built/As Installed documents submitted shall have a stamp with similar words to the following and be signed prior to submission for review.

AS INSTALLED DRAWING
We certify that this drawing is an accurate record of the work installed by our company under the terms of our contract.
Contractor's Name
Contractor's Address
.....
Signed Date
Printed

Progressively record changes to form a record of work as installed.

Prepare and submit for approval as-built drawings covering all aspects of the work.

Show dimensions, types and locations of equipment, cables, piping, ductwork pits and markers in relation to permanent site features and other underground services.

Show the 'as-installed' locations of building elements, plant and equipment with particular emphasis on items requiring maintenance or clearing.

Show off-the-grid dimensions where applicable. Include relationship to building structure street features and other services and changes made during commissioning.

Show inverts of drainage and relative locations of gully and boundary traps

Provide As Built documentation as per Schedule of Required As Builts:

6.5.3 *Operating and maintenance manuals*

- 1 Provide comprehensive Operating and Maintenance documentation to cover all installations in order to provide a detailed understanding of the plant and its operation, an aid for training of operators, a reference for fault diagnosis and a framework for preventative and breakdown maintenance.

Submit manuals in two stages. This manual shall be submitted well in advance of the Practical Completion and should be signed off by the Consultant Engineer at least 4-6 weeks before programme Practical Completion Date.

- a First draft manuals Stage 1: Submit a first draft manual 8-10 weeks before the date for practical completion for review. To include a complete manual excluding as built drawings, and other data not yet complete as part of final commissioning.
- b Final copies Stage 2: Incorporate feedback from review and from training of Principal's staff, including preparation and insertion of additional data. Include a section containing commissioning test reports. As built documentation

Revisions: Provide any revisions updates, by clearly identifying, and listing all individual inclusions, or exclusions amended from final version. This information shall be provided independently to the actual manual.

Warning and Cautions: Include to emphasise conditions hazardous to personnel or equipment, giving instructions to avoid the hazard. Format to be:

- c **WARNING:** An examining or testing procedure or practice which must be observed or risk loss of life or injury to personnel.
- d **CAUTION:** An examining or testing procedure or practice which must be followed or risk damage to equipment.

The manual shall be neatly prepared and bound in one, or a series of, vinyl hard-back folder with lettering on the front and the spine including the following information:

<u>OPERATION AND MAINTENANCE MANUAL</u>	<u>VOLUME NUMBER</u>
<u>PROJECT NAME</u>	<u>DATE OF ISSUE</u>
<u>xx SERVICES</u>	<u>CONTRACTOR</u>
<u>CONSULTANT ENGINEERS – WSP</u>	<u>SUB CONTRACTOR</u>

Pagination: Number pages consecutively. Double sided on A4 size (minimum 50%) recycled white (minimum CIE whiteness of 140) paper, with reinforced binder perforations.

Ring size: 50mm maximum, with compressor bars.

Dividers: Durable divider for each separate element, with typed description of system and major equipment components. Clearly print short titles under laminated plastic tabs.

Drawings: Fold illustrative drawings to A4 size and accommodate them in plastic pockets in the binder. Title block to be visible without removal of drawing

All aspects of the style and quality of the manual, including folders and contents shall be submitted for approval prior to completion.

6.5.3.1 Electronic/digital format

- 1 Manual shall be contained within a **single searchable PDF** accessible via Adobe or similar PDF reader. PDF shall be indexed, divided into folders and titled. Each folder shall contain relevant electronic/digital format documentation in line with the contents specified below. Refer below of a screen print of Adobe Reader for a typical arrangement.

Avoid submitting manually scanned Manufacturers' data, where possible include only searchable PDF documentation.

Include only relevant diagrams, equipment data in clear concise English. Avoid jargon and esoteric pseudonyms.

Provide each document included in the PDF with the following naming convention:

- a Section “X” – “Document Description”.pdf

Provide clear scanned copies of Statutory Certificates of Compliance

- b Accessible Dwg format CAD drawings of the as-built documentation shall be contained within the PDF.

6.5.3.2 Installation manual format

1 Section 1 – Index:

- a Index all sub-divisions of each section including lists of drawings, equipment, etc. for quick reference.
- b Include index in every volume of the manuals.

Section 2 - General Description and Capacities of Installations:

- a Divide into sub-sections for general items and for each individual system as appropriate and include full details
- b Include a directory: Names, addresses and telephone, email and facsimile numbers of Principal, Consultant Engineer, Sub-Consultants, Contractor and names of responsible parties.
- c Include a scope of works: Statement of scope of services and interfacing with other Contracts.
- d Include an installation description: General description of installation.
- e Systems description: Technical description of the systems installed, written to ensure that the Principal’s staff fully understand the scope and facilities provided. Identify function, normal operating characteristics and limiting conditions. Include schematic diagrams.
- f System performance: Technical description of the modes of operation of the systems installed.

Section 3 – Equipment:

- a Sub-divide as for Section 2 and list all major items of equipment installed complete with manufacturer's name, agent's name, contact details, model and/or type No., serial No., size, design ratings and including all relevant data necessary for re-ordering or replacing the equipment or components of same. As far as practicable, all equipment should be broken down to individually identifiable items.
- b Provide manufacturers’ technical literature for equipment installed, assembled specifically for the project, excluding irrelevant matter. Mark each product data sheet to clearly identify specific products and component parts used in the installation and data applicable to the installation. Do not include advertising literature. Cross out any section of literature which is not applicable to the project.

Section 4 – Installation, Maintenance and Operating Instructions:

- a Sub-divide as for Section 2 and include manufacturer's installation, maintenance and operating instructions for each item of equipment.
- b Emergency maintenance procedures, including telephone numbers for emergency services and after hour’s contact for suppliers and Contractors and procedures for fault finding.
- c Manufacturer’s technical literature as appropriate. Register with manufacturer as necessary. Retain copies delivered with equipment.
- d Detailed recommendations for preventative maintenance frequency and procedures which should be adopted by the Principal to ensure the most efficient operation of the systems installed. Include inspection, testing and maintenance programme in tabular form showing frequency and level of routine checks for each item.
- e Safe trouble shooting, disassembly, repair and reassembly, cleaning, alignment and adjustment, balancing and checking procedures. Provide logical step-by-step sequence of instructions for each procedure. Include calibration and recommissioning of controls.
- f Schedule of spares recommended to be held on site, being those items subject to wear or deterioration and which may involve the Principal in extended deliveries when replacements are required. Include complete nomenclature and model numbers, and local sources of supply. Include sectionalised diagrams of machines identifying component parts.
- g Schedule of normal consumable items, local sources of supply and expected replacement intervals up to a running time of 40,000 hours. Include lubricant and lubrication schedules for equipment where applicable.

- h** Add to the manufacturer's technical literature, for assembly, operating, maintenance and disassembly, safety information for all plant, systems or structures installed within the project or for existing plant, systems or structures where assembly, operating, maintenance and disassembly are altered or amended by the installation of plant, systems or structures within the project.
- i** The added safety information shall be included at relevant sections of the manufacturer's literature and not as a separate part and shall be made clear and obvious to the user of the manual.

Section 5 – Plant Operating Instructions:

- a** Sub-divide as for Section 2 and provide a complete description and correct sequence of all actions necessary to start-up, operate and shut-down all plant including procedures for seasonal changeovers where applicable. These instructions shall include full information on such items as normal and abnormal gauge readings, instrument settings and control points, differentials, time delays, oil levels, water temperatures, and all similar relevant variable and adjustable items, to permit checking and adjustments where practicable and identification of hazardous conditions or malfunction of plant. In addition, the instructions shall include information on the immediate action to be taken in the event of hazardous conditions arising.
- b** Add to the manufacturer's technical literature, for assembly, operating, maintenance and disassembly, safety information for all plant, systems or structures installed within the project or for existing plant, systems or structures where assembly, operating, maintenance and disassembly are altered or amended by the installation of plant, systems or structures within the project.
- c** The added safety information shall be included at relevant sections of the manufacturer's literature and not as a separate part and shall be made clear and obvious to the user of the manual.

Section 6 – Performance Test Results:

- a** This section shall be sub-divided as for Section 2 and shall include all performance test results as outlined in previous Clauses.

Section 7 – Certificates & Warranties

- a** Statutory Certificates of Compliance for:
- b** Electrical work.
- c** Plumbing drainage or gas-fitting work.
- d** Refrigeration or air conditioning and ventilation work.
- e** Copies of manufacturers' warranties.
- f** Certificates from authorities and utilities.
- g** Production certification.
- h** Copies of test certificates for the installation and equipment used in the installation.

Section 8 – Drawings

- a** Drawings and technical data: As necessary for the efficient operation and maintenance of the installation.

6.5.4 Warranties

- 1** Warranty periods are deemed to end at expiry of defects liability period unless specified otherwise.

Name the Principal as warrantee. Register with manufacturers as necessary. Retain copies delivered with components and equipment.

Commence warranty periods at practical completion or at acceptance of installation, if acceptance is not concurrent with Practical Completion.

If installation is not being manufactured and product warranty is conditional on the manufacturer's approval of the Contractor, submit the manufacturer's written approval of the installing firm.

6.5.5 *Required operating instructions*

- 1 Provide operating instructions and schematic diagrams mounted alongside equipment to meet statutory requirements, e.g. Fire Brigade and Worksafe. Diagrams to be colour-coded, colour-fast, laminated and mounted.

6.5.6 *Training of principal's nominated personnel*

- 1 The service of a fully knowledgeable Commissioning Engineer shall be required for a period of 1 full working week to instruct the Owner's nominated representatives in all details of the plant operation.

Conduct training at agreed time, at system or equipment location.

Immediately after Practical Completion, explain and demonstrate to the building users or nominated representative, the purpose, function and maintenance of the installations.

Use qualified personnel who are knowledgeable about the installations.

Prior to Practical Completion, explain and demonstrate to the Principal's nominated personnel the purpose, function, operating and maintenance of the installations.

Use items and procedures listed in the Operation and Maintenance manuals as the basis for detailed instruction of Principal's nominated personnel.

Conduct training at agreed time, at each system of equipment location.

Upon completion, submit certificates of training, signed by trainer and trainees.

Provide qualified manufacturer's training representatives who are knowledgeable about the installations

During the warranty period, provide technical assistance and advice to the Principal's nominated personnel regarding the operation and maintenance of the plant.

For equipment requiring seasonal operation, demonstrate during the appropriate season and within 6 months of Practical Completion.

Include copy of training material within the O&M Manual

6.5.7 *Spare parts schedule*

- 1 At least 8 weeks before the date for Practical Completion, submit a schedule of spare parts necessary for maintenance of the installation. State against each item the recommended quantity, and the manufacturer's current price, including for:
 - a Packaging and delivery to site.
 - b Checking receipt, marking and numbering in accordance with the spare parts schedule.
 - c Referencing equipment schedules in the operation and maintenance manual.
 - d Painting, greasing and packing to prevent deterioration during storage.

6.6 Post Practical Completion and Defects Liability Period

6.6.1 *Warranty and defects liability*

- 1 Warrant the whole of the Contract Works, for the Defects Liability Period (DLP), against defective workmanship and materials and against non-compliance of equipment and/or complete system with specified performance and operation.

The Defects Liability Period shall continue for a period of 12 months after the date of issue of Practical Completion and during this period, be responsible for making good on a timely manner, defects arising from the defective design materials, premature component failures, or workmanship or from any act of the Contractor that may develop in the work.

Perform maintenance during the warranty period to maintain warranty conditions. Maintenance shall comply with the current regulatory requirements.

In the event of inclusion of equipment normally covered by a lesser time warranty, or whose warranty would otherwise normally expire during the warranty period, allow for and include the costs of extending such warranty to that covering up to the end of the Defect Liability Period.

Where warranted work or equipment has failed, the warranty shall recommence on the date of acceptance of the rectified work for an additional period of time, extended by the period during which the Principal was without the use of the work or equipment. For rectified work involving new replaced equipment, the warranty for the new component shall be from date of replacement for the 12 months or longer as required by this specification

6.6.2 *General DLP maintenance scope*

1 Maintenance services shall include:

- a** Preparation of a maintenance management plan which includes a detailed maintenance programme, equipment schedules, statutory and preventative maintenance schedules, OH&S risk assessment, inspection and testing schedules, breakdown rectification management and reporting systems. Submit maintenance plan and schedules prior to Practical Completion.
- b** Preventative and corrective maintenance and testing services to assure specified system reliability and availability.
- c** Statutory maintenance and testing required by the Building Regulations and other authorities.
- d** Breakdown and system fault responses. Attend site, ascertain cause of alarm and rectify where possible. Make system alterations at own cost to prevent future system faults. Pay any Authority charges resulting from callout on system faults.
- e** Statutory regular system fire and life safety system control and performance verification test to be carried out after hours.
- f** Comply with Building and Occupational Health and Safety Regulations. Comply with manufacturer's recommended maintenance programme and schedules.
- g** Monthly maintenance and servicing in accordance with maintenance schedules recommended by equipment manufacturers.
- h** Rectification of defects at Contractors cost. Replacement of consumables at Contractors cost during the maintenance period.
- i** Submission of monthly reports to Principal and Consultant Engineer on maintenance activities performed.
- j** Provision and filling out log book which must be kept on site, detailing all visits.
- k** Service personnel to be available for call out on a 24 hours per day, 7 day per week basis. Response time for emergency and corrective maintenance must not exceed 2 hours.
- l** Operation and verification in conjunction with other services trades, of all fire and life safety systems. Such testing to be undertaken at 6 monthly intervals.

6.6.3 *Final completion*

- #### **1 Final Completion will not be granted until maintenance records are complete and signed off and until all defects have been rectified and inspected.**

The Maintenance and Defects Liability Period may be extended unless maintenance and servicing visits have been made and reported as specified.

6.6.4 *Submissions*

1 Provide:

- a** Maintenance management plan.
- b** Monthly Maintenance reports.

c Completed maintenance records

Mandatory maintenance records to include monthly maintenance and service reports, including water treatment reports, reports on all defects and their rectification, certification of the correct operation of essential services, a maintenance log book and any other records which must be kept in order to comply with legislative or OH&S requirements.

Monthly Maintenance and Service Reports to include:

- d** Check list of all items serviced and inspected.
- e** Description of maintenance performed.
- f** Notes on supplementary maintenance required.
- g** Comprehensive water treatment report.

Maintenance Log Book to include:

- h** Schedule of all equipment.
- i** Schedule of all control sensors and control set point values.
- j** Register of maintenance visits and cross reference to service reports.
- k** Register of work performed under headings of preventive maintenance, corrective and emergency maintenance and defects rectification.

The Maintenance Log Book is to be kept in a locked cabinet provided for the purpose in an agreed location on site.

At the conclusion of each maintenance visit the log book is to be duly completed and signed off by an authorised representative of the Principal /Building Owner and service records are to be forwarded to the Principal /Building Owner or his representative within 3 days of each visit.

6.6.5 DLP preventative maintenance and testing

6.6.5.1 General

- 1** As a minimum, carry out all maintenance and servicing recommended by the manufacturer of each piece of equipment at the recommended intervals.

6.6.5.2 Hydraulic services requirements

As a minimum, regular maintenance and servicing tasks to include.

Monthly:

- a** Check all water storage tanks are water tight and clear all floating debris from water surface.
- b** Check and report on necessity for replacement of filter media.
- c** Check all water tank isolation valves are operational and in good working order.
- d** Check all water & gas meters are operational and monthly readings are within seasonal limits.
- e** Clear as necessary drip trays, strainers and other sediment collecting items.
- f** In large toilet or shower areas check for sewer smells, ensure all traps are being primed and sanitary vents are in good working order.
- g** Check reduced pressure zone valves are operational and complete a test discharge of water.
- h** Check all hot water heaters are in good working order, complete a test discharge of the pressure relief valves, ensure safety tray are in good working order.
- i** Check hot water temperatures from basins are within limits and adjust where required.
- j** Check all stormwater downpipes are intact and water tight.

Quarterly:

- k** Check all pump control panels are clear of dust and debris. Test all BMS outputs are operational by changing over pumps or running test routines. Ensure audible alarms and strobes are clear and visible. Check all building area isolation valves are in good working order. Including isolation valves to major toilet blocks or riser connections.
- l** Check and adjust boilers, burners and water levels in water heaters.
- m** Check plumbing pipework within walk in services ducts, including wall hung cisterns, urinal flush valves and priming devices.

Six monthly:

- n** Check all roof sections, sumps and gutters are clear of organic debris and fines. Jet blast clean and repair all damaged over flashing and
- o** Check all grease interceptors and settling tanks have lids and frames secured and no stormwater ingress.
- p** Check all gully traps and grates are secure and jet clean as required
- q** Check all manhole lids and stormwater water headwalls are clear of debris
- r** Check all external hose taps are not leaking and have backflow device fitted where required.

Last maintenance visit at 12 Months:

- s** Complete all 6 month checks
- t** Check test and report on the building recycling water system including inspection of the rainwater tank
- u** Complete Hot water warranty testing as noted in manufactures warranty requirement.
- v** Replace all filter media disposable filters and UV lamp.
- w** Flush all WC's and check for cistern leaks, test all sanitary fixtures with hot and cold water flush
- x** The date for the final maintenance visit to be advised 7 days prior to the proposed visit so that a representative may be present to inspect the equipment when end covers are removed.
- y** Check and de-sludge basement seepage pump station.

7 Sewer/Sanitary Drainage Systems

7.1 General

- 1 Make the connection to the local authority's connection branch in an approved manner and to local inspector's requirements.

Ascertain the depth, position and suitability of the sewer connection points prior to the commencement of any work and incorporate any adjustments required to execute the work. No claims for redundant work will be considered due to failure to comply with this requirement. Make connection to the property drainage connection branch.

Pipelines shall be laid true to line and bore from point to point.

Unless otherwise indicated on the drawings pipelines shall be graded in accordance with the Authorities requirements and as specified under "Excavation & Pipework Requirements" section.

Supply and install an overflow gully to provide a safe release from the connection point. Terminate gully top to comply with current authorities' regulations.

Provide a hydrostatic test to choke level for a minimum period of 15 minutes and as required by the testing authority.

Provide and install clear-out inspection fittings to provide rodding access to all lines. Riser from pipeline shall extend vertically to finished floor level and terminate with a brass screw out and frame.

Supply and install all reflux valves in the locations indicated or as required by the local authority and as specified in the "Materials" section

Carry out all necessary concrete encasing as specified in "Excavation & Pipework Requirements" section and as shown on drawings

Supply and install all basket traps in the locations indicated on the drawings or as required by the local authority and as specified in the "Materials" section.

8 Sanitary Plumbing System

8.1 General

- 1 Sanitary plumbing works above ground shall include all those works generally considered by authorities and trade practice to be soil, waste, vent above ground as distinct from drainer's work.
 - a Terminate all vents in accordance with the documentation, local authority requirements and Australian Standards. Flashing of vents, which penetrate the roof, shall be carried out within the hydraulics scope of works.
 - b Install inspection openings in accessible locations so that each section of pipework can be cleaned. Inspection opening sizes shall be in accordance with authorities' requirements. Install testing gates on all stacks.
 - c Supply and install approved expansion joints to all soil, waste, relief and main vent lines. Expansion joints shall be installed on each line at intervals not exceeding the requirements of Australian Standards and on either side of building expansion joints.
 - d Traps shall be of the same material as the branch drainage except traps in visible locations shall be chrome plated. Refer "Materials" section for specified pipework.
 - e Unless otherwise noted all pan collars to soil type fixtures shall be of approved PVC materials and pattern. Collar shall incorporate an approved neoprene rubber ring joint, and shall not be offset any greater than 20mm.
 - f Ensure all drainage pipework located over retail, residential, hotel or future retail tenancies is acoustically insulated.
 - g Use trap with 100mm riser and 50mm or 65mm outlet of self-cleansing pattern.
-

8.2 Fixture Traps

- 1 75mm water seal traps shall be provided for the following fixtures:
 - a Sinks: 50 mm two-part universal pattern.
 - b Basin: 40 mm two-part S or P-trap with 40 mm extension riser.
-

8.3 Insulation of Cold Waste Piping

- 1 All waste pipes receiving discharge from cool rooms, FDC units and refrigerated cabinets which pass through areas where condensate forming externally on the pipework may cause damage such as in multi-level buildings and car parks shall be insulated.

The insulation material shall be low temperature pipe insulation or approved equal with aluminium foil outer sheathing and shall be fully vapour sealed at all joints with aluminium adhesive tape strictly in accordance with the manufacturer's recommendations.

9 Stormwater Drainage and Downpipe Systems

9.1 General

- 1 Make the connection to the local Authorities stormwater in an approved manner and to local Authorities requirements. Ascertain the depth, position and suitability of the stormwater connection points prior to the commencement of any work and incorporate any adjustments required to execute the work.

Pipelines shall be laid true to line and bore from point to point.

Provide inspection pits at changes of direction. Pits to be reinforced concrete of appropriate dimensions to Code with depths to match invert levels of pipeline.

Gutters, downpipes and overflows shall have adequate capacity for 100 year recurrence interval storms of 5 minutes duration.

All new gutters to be fitted with gutter guards.

All eaves gutters to overflow away from building envelope/façade.

Downpipes to terminate 150mm above finished surface level and discharge over grated pits.

Provide roof and balcony drainage outlets in the positions indicated on the drawings and as scheduled. Each outlet shall be cast into the concrete slabs complete with rubber ringed inverted adapter and connection to the vertical downpipes. Support each sump during construction with wire ties, set level both ways, to suit roof falls.

The overflow system shall either directly discharge through the external walls or interconnect via a dedicated piping system then discharge independently of the main stormwater runs.

Downpipes cast in columns shall be HDPE. Internal downpipes shall be provided with test gates at the base to enable full testing. Flexible rubber ring joints are to be provided where the downpipe connects to the box gutter outlet.

Test gates shall be provided at the base of all internal downpipes and of the same size as the downpipes prior to connecting to stormwater drainage lines. Inspection openings generally shall be provided at every junction, bend and change of direction.

Supply and fix approved, flat or domed type roof drainage outlets cast in the concrete roof structure in positions indicated on the drawings. Set grates, flush and matching with adjoining surfaces. Grates shall have puddle flange and membrane clamp.

Where a branch line enters a vertical pipe, the branch fitting must be wholly outside the vertical and horizontal pipe, such that the internal bore of the pipe is maintained at all times. All horizontal and vertical branches shall be made at 45° to the main line.

9.2 Sub-soil Drains

- 1 Provide subsoil drains to intercept groundwater seepage and prevent water build-up behind walls and under floors, pavements and landscape beds. Connect subsoil drains to surface drains or to the stormwater drainage system as applicable.

Follow the minimum clear depths, measured to the crown of the pipe, where the pipe passes below the following elements:

- a 100mm below formation level of the pavement, kerb or channel.
- b 100mm below the average gradient of the bottom of footings.

- c Below the finished surface of unpaved ground, levels as indicated on drawings.
- d Minimum trench width to be 300 mm.

Bed piping on a continuous underlay of bedding material, at least 75 mm thick after compaction. Lay the pipe with one line of perforations at the bottom.

If necessary, form chases to prevent projections such as sockets and flanges from bearing on the trench bottom or underlay.

Place the material in the pipe surround in 200mm layers lose thickness, and compact without damaging or displacing the piping.

Provide polyester permeable filter socks capable of retaining particles of 0.25 mm size. Securely fit or join the sock at each joint.

10 Domestic Cold Water Systems

10.1 General

- 1 Locate valves behind access panels or within accessible ducts. Valves located in ground shall be provided with CI path box and lid set into concrete surround with pipe riser around valve stem.
- 2 Supply and install control valves to each group of fixtures. Locate behind access panels or within accessible ducts.

Provide unions at wall or floor surface and at fixtures and appliances to allow removal and replacement without the need to adjust connections.

To all changes of direction on rubber ring jointed pipelines below ground install concrete thrust blocks to restrain the internal operating pressures of the pipeline under all conditions. Concrete mass shall be poured around and behind fittings and bear against virgin soil material. A minimum of 0.75 cubic metres of concrete shall be used at each position.

Unless indicated otherwise taps and faucets shall be all brass construction. All connections and cover plates shall be bright chromium plated finish. Taps shall be fitted with anti-splash nozzles, except for hose taps and/or where otherwise specified.

Supply and install an approved BFPD in the domestic cold water supply and irrigation system as required by the Water Authority and as specified in the “Materials” section. Include maintenance instructions in as-built manuals to be supplied at end of project establishing an authorised maintenance programme including registration and certification of all installed devices. Provide isolation valves upstream and downstream of valve and line strainer at inlet to BFPD. Flush piping before installing device and test device after installation and prior to operation in service.

Provide and install hose taps with anti-vandal heads. Each tap shall consist of a 20mm diameter copper riser with a back plated elbow and a 20mm diameter brass finish hose tap. The tap shall be 450mm above the floor unless otherwise directed.

Provide a water pressure test of 1500 kPa for a period of two hours. Disconnect any equipment connected to the service not rated to the test pressure, before testing commences.

10.2 Sub-metering

- 1 Approved sub-meters to be installed in accordance with Rockhampton Regional Council and CQU requirements.

10.2.1 Authority

- 1 Approved sub-metering is to be installed in accordance with Rockhampton Regional Council policy.
- 2 Provision for wireless Automatic Meter Reading [AMR] technology and BMS connectivity shall be provided, and all meters shall be readable through the AMR and BMS system.

11 Domestic Hot Water and Warm Water Services

11.1 General

- 1 Provide unions at wall or floor surface and at appliances to allow removal and replacement without the need to adjust connections.

Supply and install control valves to each group of fixtures. Locate behind access panels or within accessible ducts.

Make adequate provision for expansion and contraction by the provision of clips/brackets with wooden expansion blocks so that under all working conditions no strain is imposed on pipework or fittings. Pipes located in walls and floors shall be provided with sufficient insulation so that expansion and contraction does not impose a strain on the pipework or finished surfaces. No joints will be allowed within or under concrete slabs where slab is on ground.

Branch pipes off straight lengths of unrestrained supply pipes shall incorporate a minimum of 3 long radius pipe bends and a straight length of pipe not less than 1000mm long between the branch connection and the first branch pipe restraint to facilitate movement of the supply pipe without imposing strain on the pipe connections and fittings.

11.2 Hot Water Heaters

- 1 Provide heaters with the following:
 - a An automatic surface mounted temperature control thermostat with adjustable ranges on automatic over-temperature cut out to shut down the fuel or power supply should the thermostat malfunction.
 - b A temperature and pressure relief valve to conform to the requirements of AS 1357 and terminate over a tundish or FW as noted on drawings.
 - c Electric hot water heaters shall be in accordance with AS 1056 and be tested and approved by the electrical supply authority.
 - d A safe tray shall be supplied and installed under the hot water heater. A drainage outlet shall be fitted and sealed to the tray and discharge as noted on the drawings. The hot water unit shall be no closer than 20mm from the side of the trays.
 - e Hot water heaters shall be spaced from the mounted surface with hardwood battens or proprietary hot water unit “sandwich” support.
-

11.3 Backflow Prevention

- 1 Provide approved BFPDs in the domestic hot water supply as required by the local Authority and as specified in the “Materials” section.

Provide isolation valves upstream and downstream of valve and line strainer at inlet to BFPD.

Flush piping before installing device and test device after installation and prior to operation in service.

11.4 Thermal Insulation

1 Provide all necessary insulation to hot water piping to prevent heat loss.

Insulation shall not be installed until all relevant tests and inspections have been carried out.

The pipe insulation shall comprise closed cell polyethylene foam and shall be installed around the pipework surface as tightly as possible without gaps, with the edges and ends tightly butted together.

All joints shall be taped using a 75mm adhesive backed PVC tape or, if faced with aluminium foil, joints shall be taped using a 75mm pressure sensitive aluminium tape. Where necessary the preformed sections shall be cut (using a sharp knife or scissors) and mitred, to ensure a tight fit around bends and at tees. Joints in insulation and at brackets, bends, along longitudinal joints and elsewhere at 500mm centres shall all be taped as described above.

Appropriate UV protection shall be included for all thermal insulation that may be exposed to UV light. The approved protection shall be in accordance with the manufacturers requirements.

At union flanges, valves and strainers, or any areas where access is required for maintenance; insulation shall incorporate the press seal fastener system.

Where supports and spacer blocks are used to support the pipework, the pipe insulation shall be butted tightly up to the spacer blocks and sealed using a silicon sealant.

Apply insulation in accordance with manufacturer's instructions in a neat and workmanlike manner so as to present a smooth even finish.

Provide timber blocks at brackets to ensure even support without affecting insulation.

Insulation to piping chased into walls to be 10mm Ableflex or approved equal. Ensure that such insulation permits adequate expansion and contraction.

12 Re-use Water System

12.1 General

- 1 Provide unions at wall or floor surface and at appliances to allow removal and replacement without the need to adjust connections.
- 2 Supply and install control valves to each group of fixtures. Locate behind access panels or within accessible ducts.

Make adequate provision for expansion and contraction by the provision of clips/brackets with wooden expansion blocks so that under all working conditions no strain is imposed on pipework or fittings, Pipes located in walls and floors shall be provided with sufficient insulation so that expansion and contraction does not impose a strain on the pipework or finished surfaces. No joints will be allowed within or under concrete slabs where slab is on ground.

Branch pipes off straight lengths of unrestrained supply pipes shall incorporate a minimum of 3 long radius pipe bends and a straight length of pipe not less than 1000mm long between the branch connection and the first branch pipe restraint to facilitate movement of the supply pipe without imposing strain on the pipe connections and fittings

Balancing valves shall be all bronze globe valves with limited stop in the open position. All other valves must be bronze.

12.2 Rain Water

- 1 Rain water shall be automatically switched to potable water when water storage tank levels are depleted, and automatically switched back when storage tank levels are restored.

All switching shall be via level sensors installed in storage tanks.

12.3 Filters

- 1 Supply and fix filters in the locations indicated on the drawings.

Each filter shall have a back-up filter of 100% capacity has to be installed parallel to it.

Provide filters with the following:

- a An automatic differential pressure control with adjustable ranges on automatic switch over to second filter.
- b Pressure gauge of 100mm dial in kPa.

13 Natural Gas

13.1 General

- 1 Install regulators in the positions indicated on the drawings and under and over pressure shut-off regulator with the capacity to pass the required amount of gas reducing the pressure as required. Regulators are to be the OPSO type and be capable of delivering the required megajoule rating and be vented to the requirements of the Gas Authority and AS5601.1.

Allow to purge the pipework in accordance with standards.

Gas pipework shall maintain a gradient of 1:230 unless otherwise indicated or directed by the gas authority. The graded pipes are to grade to low points terminating in a tail pipe with a 20mm capped valve in an approved easily located and nominated position.

Provide all gas piping, appliance or equipment gas control valves, gas appliance regulators, stainless steel flexible pipe connections, gas unions and ancillary fittings for all gas burning appliances and other equipment or machines wherever located in the building to ensure the full operation of these items of equipment.

Before connecting gas to gas heaters, appliances and equipment ensure that the appliance is approved in accordance with the requirements of the Qld Gas Examiner and is correctly installed and that all supports, flues, connections, flame failure devices comply with the regulations.

Gas leakage detection shall be installed. Detection warning shall be alarm and/or interconnected with a gas shutoff device.

Gas pressure proving shall be installed. Leakage warning shall be alarm and/or interconnected with a gas shutoff device.

13.2 Gas – Systems Shut Off

- 1 Supply and install multiple automatic shutdown control valves, as approved and required by the local Authority and the project Fire Engineer.

Automatic shutdown equipment shall incorporate valves, motorised valves, pressure switch, flanges and other items of equipment as necessary. Connect control wiring from the emergency power supply for power operation of motorised valves. The whole equipment shall shut down the gas supply during a fire event and must be reset manually before gas supply can be restored.

Provide a set of voltage free alarm contacts to the Gas Service Automatic Shutdown valve for connection to the BMS System. The Shut-Off system is to be similar to those manufactured by SCE Systems 3.

Each kitchen >100m² shall have a localised emergency shut down button and associated wiring (both by Electrical Contractor) to send signal to a localised gas solenoid valve (supplied and installed by Plumbing Contractor) installed to locally shut off gas supply to the immediate downstream kitchen.

Gas shutdown interlock shall be provided for kitchen exhaust hoods.

13.3 Gas Pressure Regulators

- 1 All gas regulators shall incorporate over pressure protection (over pressure shut off - OPSO) built in.
- 2 Regulators shall be suitable for the expected megajoule peak flow, and suitable to meet the incoming and outgoing pressure requirements.
- 3 All gas regulators shall be vented to atmosphere as required in accordance with AS5601.1.

13.4 Enclosures

- 1 All enclosures shall be ventilated in accordance with AS5601.1.
- 2 There is to be no services installed in enclosures other than services associated with the gas system.
- 3 Conduits or open pipes entering the enclosure are to fully sealed and airtight to prevent gas from entering the conduit.
- 4 Ceilings walls and floors shall be fully sealed to prevent gas from entering adjacent spaces.

13.5 Valve Boxes

- 1 Supply and install valve boxes for access to underground valves. The boxes shall be standard cast iron boxes with hinged covers. Set tops flush with pavement surfaces, or 15 mm above unpaved surfaces. Provide an in situ concrete surround 150 mm thick, to the top, bottom and sides of the box, with exposed surface steel trowelled. Set a shaft of pipe beneath each box to give access to the valve. Stamp the box cover with the word "GAS".

13.6 Wall Boxes

- 1 Provide wall boxes to accommodate above ground valves, regulators and the like. The boxes shall be constructed of 1.2mm stainless steel plate continuous welded box construction with leading edge folded twice at 90° to form 25 x 25mm frontal surround. Fix to masonry backing with four 10mm galvanised masonry bolts. Set the bottom of the box to fall outward. Form four 10mm diameter holes in the frontal surround section at box floor level. Provide, to the box floor inlet and outlet pipes, sleeves formed of 1.2mm thick steel tube with 1.2mm galvanised flanges to pipe diameter plus 50mm. Bed each flange on epoxy mortar and rivet to the floor of the box with four 3mm diameter rivets.

Provide metal frame doors with 2.5mm clear float glass glazing. Affix to the glass with adhesive a white laminated plastic label 200 x 100mm, engraved with red letters "IN CASE OF EMERGENCY BREAK GLASS AND SHUT VALVES". Provide lock, keys and two 100mm brass hinges.

13.7 Gas Services Risers

- 1 All gas pipework risers shall be installed in pipe riser shafts containing gas services only. The services ducts/shafts shall be ventilated at top and bottom in accordance with AS5601.1 appendix K, and shall form a single separate fire compartment. Allowances shall be made for adequate airflow and ventilation throughout the entire riser shaft.

Horizontal penetrations shall be fire protected to ensure separate compartmentation is maintained in accordance with the NCC.

13.8 Gas Vertical Pipe Support Systems

- 1 Gas pipework shall be installed in accordance with AS5601.1 appendix K sections K7, K8, and K9.

14 Tanks

14.1 Rain Water Harvesting Tanks

Shall have the following minimum requirements:

- 1 Tank structural foundations (or similar support) to be capable of supporting the tank to full overflow capacity with a safety margin. Design to be by a competent qualified Structural Engineer
- 2 Total effective storage capacity shall not include upper freeboard, and lower freeboard (prevent air suction and pump cavitation).
- 3 Upper freeboard to accommodate water level above overflow during peak design inflow
- 4 Tanks within the building footprint (or where overflow can enter the building) shall have an orifice plate fitted to the inlet to limit the maximum possible inflow during any rain event, and shall cater for an flooded inlet pipe network. The overflow system (from tank) shall cater for the inlet flow rate derived accordingly including a 10% safety factor. The overflow network shall cater for this full flow and any additional design flows to the point of discharge outside of the building footprint.
- 5 An overflow shall be provided for each tank and shall discharge to atmosphere with appropriate vermin protection (frog flap).
- 6 Sludge valves shall be provided to each tank half and shall be minimum 0.5 times the ID of the tank suction pipework. The internal tank floor shall be graded to the invert level of the sludge valve outlet
- 7 Access lids to be sealed to prevent dust/dirt/debris from entering the tank.
- 8 Electronic water level controls.
- 9 Tanks shall have low-level pump deactivation switch connected to BMS.
- 10 Signage shall be provided fixed to the tank wall indicating:
 - a Effective Storage Capacity.
 - b Tank use.
 - c Confined space warning (if applicable).
- 11 Pump suction (where applicable) shall have vortex plate in tank to reduce the lower freeboard volume.

14.2 Tank Water Level Controls

Provide a system of water level controls to the storage tanks, to automatically operate the pumps and activate the alarms, including wiring/connection to the pumps panel.

The water level controls shall be non mechanical sensing type.

The sensor shall be mounted within the tank where it remains accessible and has an uninterrupted reading of water levels over the entire height of the tank.

Functions shall include:

- 1** Low Level:
 - a** Pump Stop.
 - b** Mains Water Diversion Valve Actuating.

Provide all control wiring between pumps in a weather-proof enclosure, and water storage level controls. The system shall be installed in accordance with the manufacturer's instructions and Authorities requirements. A 240-volt power supply shall be provided by others.

15 Pumps

15.1 Sanitary Drainage Pump Stations

Provide a sanitary drainage pumping system, generally comprising:

- 1 Identical submersible pumps.
- 2 Controls and wiring.
- 3 Rising main.
- 4 Chamber vent.
- 5 Integrated valve box with independent air tight lid
- 6 Holding well with independent air tight lid.
- 7 Control panel, level sensors/controllers and wiring to give automatic operation of pumps.
- 8 Hose tap to be provided with BFPD within 10m of pump station access lids.
- 9 Site commissioning by trained Service Personnel after installation by others.

All to form a complete and approved installation as indicated on the drawings.

15.1.1 *Pumps to Comply with the following Requirements*

- 1 Submersible type with shredding/grinding impeller.
- 2 Have lubricated for life, maintenance free motor shaft bearings and high quality silicon carbide mechanical seal independent of direction of rotation; lip seal oil lubricated.
- 3 Have waterproof cabling.
- 4 Shall be auto-coupling (to fixed pipework), on fixed stainless steel guide rail(s)

15.1.2 *Pump Well Body to Comply With The Following requirements*

- 1 Incorporate emergency storage volume (assuming both pumps are stopped) where applicable and nominated within the documentation and schedules.
- 2 Chamber body to have provisions to allow for anti-buoyancy retainment.
- 3 If not structural, shall be of fibre reinforced plastic or equal performing material.
- 4 Valve box, and pump access lids shall be rated for relevant trafficable conditions.
- 5 All pipe penetrations to be fully sealed and waterproof with threaded couplings or similar.
- 6 Valve box to contain manual bleed/tapping point
- 7 Valve box to have drain to main chamber with check valve
- 8 Pump operation/switching shall be by automated sensors mounted within pump well. Level switching to operate;
 - a Pump Stop
 - b Duty pump start
 - c Standby pump start
 - d High level alarm

15.1.3 *Pump Well Water Level Controls*

- 1 Provide a system of water level controls to the pump well, to automatically operate the pumps and activate the alarms, including wiring/connection to the pumps panel.
- 2 The sensor shall be mounted within the well where it remains accessible and has an uninterrupted reading of water levels over the entire height of the tank.
- 3 Functions shall include:
 - a Activate High Level Alarm.
 - b Stop Duty Pump.
 - c Start Duty Pump.
 - d Stop Standby Pump.
 - e Start Standby Pump.
- 4 Provide all control wiring between control panel, and level controls. The system shall be installed in accordance with the manufacturer's instructions and Authorities requirements.

15.1.4 *Control Panel*

Provide control panel, level controllers and wiring to give automatic operation of pumps.

Control panel to be located where shown on plans, fixed to wall or freestanding and shall be suitable for external installation exposed to weather conditions. Location to be confirmed prior to installation.

For control panel requirements refer to "Pump Control Panels" section.

15.2 Domestic Cold Water / Grey Water / Rain Water Booster & Pressure Pumps

15.2.1 *Pumps to Comply with the following Requirements*

Pump set is to incorporate the following:

- 1 Multiple vertical variable speed displacement pumps.
- 2 Separate distribution board fitted with main power supply isolating switch with separate circuit breakers for each unit.
- 3 Inlet side of each pump pre-plumbed with an isolating valve and non-return valve, connected to a common inlet manifold with a neoprene vibration eliminator.
- 4 Discharge side of each pump fitted with an isolation valve and a common manifold is to be provided with a separate main isolating valve, neoprene vibration eliminator and a full flow bypass. On the discharge manifold 65mm pressure gauge is provided together with a single pressure tank in accordance with the manufacturers specification. This system will maintain a set pressure with a maximum variation of 2.0%.
- 5 Site commissioning by trained Service Personnel after installation by others.

15.2.2 *Control Panel*

Provide control panel, tank level controllers (for tank suction pressure pump sets) and wiring to give automatic operation of pumps.

For control panel requirements refer to "Pump Control Panels" section.

15.3 Hot Water Circulating Pumps

15.3.1 *Pumps to Comply with the following Requirements*

- 1 Provide and install pressure gauges and gauge control valves on the inlet and outlet of each pump.
- 2 Fit a “Gate” valve on the inlet of each pump and a balancing valve on the outlet of each pump.
- 3 Provide stainless steel flanged vibration eliminators on the suction and delivery connection to each pump. Provide stainless steel vibration eliminators on the pipework connection to and from the pump set group.
- 4 Provide air eliminators on each side of the pump.
- 5 Provide and install anti-vibration mounts as required.
- 6 Provide electrical power, control and alarm wiring with conduits from the pumps electrical control panels to pump motors. Pumps to have thermal overload protection auto/off/manual selector for each pump, auto/change-over selection switch, motor, circuit breakers, AC on light, pump run lights, pump fail lights, alarm button and mute button.
- 7 Electrical wire each pump for 24 hours running with an automatic change over to the second pump after each 24-hour duty.

15.3.2 *Control Panel*

Provide control panel, and wiring to give automatic operation of pumps.

For control panel requirements refer to “Pump Control Panels” section.

15.4 Pump Control Panels

Supply and install all equipment necessary to operate the pumps specified under hydraulic services.

Allow for all control cabinets, mounting brackets, contactors, isolating and control switches, auxiliary switches, alarms, wiring between pump and panel, panel and level controls, and other associated equipment necessary for the safe and effective operation of the pumps as required for the installation and in accordance with statutory requirements to AS 3000.

Provide a complete specification and drawings of pumps and control equipment prior to installation.

15.4.1 *Functions*

15.4.1.1 All Control Panels

Each control panel shall contain the following minimum.

- 1 Circuit breakers for each pump.
- 2 Control circuit breaker.
- 3 Auto/Off/manual switch to each pump.
- 4 Automatic alternation of pumps.
- 5 Audible alarm complete with reset type mute button.
- 6 Individual digital hour meters for each pump operation.
- 7 All switches to be fitted in inner door to prevent unauthorised usage.
- 8 Individual digital hour run meters for each pump operation.

15.4.1.2 Pump Station Control Panels

- 1 Indicating lights for the following:
 - a power on 1 off
 - b pumps run 1 off for each pump
 - c pump fail 1 off for each pump
 - d high level alarm 1 off.
- 2 Volt free terminals on separate terminal strip for each pump run, each pump fail, high level alarm, wiring to remote indication panel by other nominated Works Packages.

15.4.1.3 Pressure Pumps Control Panels

- 1 Indicating lights for the following:
 - a power on 1 off
 - b pumps run 1 off for each pump
 - c pump fail 1 off for each pump
 - d low storage tank level (will stop pumps) 1 off
 - e low suction (will stop pump) 1 off.
 - f low discharge alarm (will stop pumps) 1 off.
- 2 Volt free terminals on separate terminal strip for each pump run, each pump fail, low storage tank level, low suction, low discharge, wiring to remote indication panel by other nominated Works Packages.

15.4.2 Control Panels – Construction

- 1 Control panel shall be provided to house electrical switchgear associated with pumps, and any other equipment requiring electrical controls for proper operation and safety. The control panels shall be wall mounted adjacent to equipment or in location shown with top at approximately 1800mm high above floor. In some cases, approval, will be given for control panels to be mounted on equipment.
- 2 Control panels shall be totally enclosed cabinet type.
- 3 Control panels shall be designed for ease of access to all equipment and wiring and shall be provided with at least 10% spare space for the addition of future equipment.
- 4 Control panels shall comply with AS 1339 as follows:
 - a IP50 for internal locations.
 - b IP54 for external locations or where a weatherproof cabinet or enclosure is specified.
- 5 Doors shall have concealed hinges and an approved key operated lock.
- 6 Two locks for doors in excess of 1000mm.
- 7 Vertical locking bars to provide top and bottom fastening shall be provided on all doors exceeding 1500mm in height.
- 8 A dustproof seal consisting of a substantial mould neoprene rubber strip shall be provided for doors or removable panels and covers. Escutcheon panels need not be provided with seals.
- 9 Switches, indicators, etc. to which wiring is connected shall not be fitted on removable plates.
- 10 Where conduits or cables enter cabinets, (cables must enter from the bottom), drill the cabinet and locknut the conduits or the cable glands to the cabinet. Note: No top or side entry without approval.

15.4.3 *Equipment*

- 1 DIN standard rail mounted terminal strips shall be provided for the termination of all external control circuit wiring and BAS identification.
- 2 Equipment shall be neatly arranged and readily accessible for maintenance.
- 3 Each control panel shall be provided with an isolating switch, neutral and earth link.
- 4 Control switches shall be rotary type complying with AS 1431.
- 5 Indicator lights shall match control switches with colours as follows:
 - a RED Pump fail, alarm – e.g. High and low level.
 - b AMBER Pump running.
 - c GREEN Power supply on.
- 6 Circuit breakers shall be approved by Australian supply authorities and comply with Australian Standards.
- 7 Time clocks shall be quartz crystal oscillator type with an accuracy of plus/minus 5 minutes per annum and complete with nickel cadmium battery to provide a 50-hour reserve/digital 100 hour.
- 8 Covered terminal strips with labelled terminals shall be provided for connection of the Building Management System (BMS) wiring by others. Wiring to terminals shall provide voltage free contacts for all alarms and faults and for pump running indication. Contacts to be either normally open or normally closed contacts.

15.4.4 *Labelling*

- 1 All control panels and equipment shall be labelled with engraved laminated type labels. Labels shall be black and white lettering. Size of lettering shall match importance and location. Labels shall be screw fixed.
- 2 Cabinets shall be labelled to indicate function
- 3 Equipment shall be labelled to indicate function and to identify any associated equipment e.g.:
 - a “PUMP CONTROL SELECTOR”
 - b AUTO/OFF/MANUAL.

15.4.5 *Testing*

Test the system on completion. Furnish two copies of the manufacturer’s test certificate.

15.5 Pump Controls Via Water Level Controls

Provide a system of water level controls to the holding wells, to automatically operate the pumps and activate the alarms, including wiring/connection to the pumps panel.

The water level controls shall be non mechanical sensing type.

The sensor shall be mounted within the tank where it remains accessible and has an uninterrupted reading of water levels over the entire height of the tank.

Functions shall include:

1. Stop Duty Pump.
2. Start Duty Pump.
3. Stop Standby Pump.
4. Start Standby Pump.

Provide all control wiring between pumps in the pump room, and water storage level controls. The system shall be installed in accordance with the manufacturer's instructions and Authorities requirements.

15.6 Flexible Pump Couplings

Flexible connections or anti-vibration couplings are to be installed on the suction and discharge side of each pump in accordance with the manufacturer's instructions. Each flexible connector shall have a hose length of not less than four (4) times the pipe diameter installed with a maximum angular deflection not exceeding five (5) degrees.

15.7 Pump Mountings

Pump mountings shall be spring isolating mountings of an appropriate size to provide a minimum 95% vibration absorption.

16 Piping, Valves and Fittings

16.1 General

- 1 Supply piping in clean, full standard, straight lengths, free from any defects, mill scale, rust, burns, dents and kinks and protected against mechanical damage.
-

16.2 Dimensions on Site

- 1 Be responsible for taking all dimensions on site. The dimensions must be checked before work is commenced or prefabricated. All levels and dimensions of existing services must be confirmed before commencing work. The invert levels shown on the drawings are recommended only and must be checked on site before excavation or installation of pipe work to ensure connection to supply sources and fall to pipe work etc. is correct.
-

16.3 Fixture and Fixture Points

- 1 Install and supply all materials, backing plate and noggins necessary to make all fixtures and fixture outlet points rigid (i.e. sanitary fixtures and tap fittings etc.). All fixtures shall be connected via a service pipe and union type fitting. Nut and olive connection is not acceptable. Direct connection of pipe work to the fixture will not be accepted.
-

16.4 Pipes in Slab

- 1 Under no circumstances shall any pipe work be cast in slabs, unless clearly noted on the drawings. Any hot and cold water pipes noted as being 'in slab' shall be contained in UPVC/polyethylene conduits and these pipes shall be totally retrievable.

The HDPE cast in concrete sanitary plumbing system shall be cast in without any sleeving or wrapping in accordance with AS 2033 and the manufacturer's requirements.

16.5 Setting Out

- 1 Set out and identify all trenches, pits, pipelines and plant equipment positions.

Gain approval for the proposed location of the works prior to commencement of any excavation.

Larger scale drawings are to be used in preference to small scale where no dimensions are shown.

Architectural dimensioned drawings are to be used in preference to scaled drawings regardless of their relative scale.

a Trenches:

- i Interpret all levels and grades to determine correct falls to pits and street connections. Advise if grades cannot be met so that revised locations or grades may be determined for the works.
- ii The set out of piping, pits and covers shall be arranged in conjunction with other trades so as to be readily accessible for operating, servicing, maintaining and repairing of it.
- iii Under no circumstance shall the Authorities and/or the Contractor conceal any drainage pipes etc. before inspection and approval.
- iv Piping, pits, etc. which are installed in unsuitable locations shall be removed and relocated as directed by the superintendent. All expenses caused by this action shall be made good without extra charge.

b Core Holes:

- i Set out all core holes and openings as required and necessary for the passage of pipes and/or conduits throughout the existing structure. Arrange for approval from the Structural Engineer to drill such holes or openings and pay all cost involved where required in existing slabs.
 - ii Core holes within "in situ" slabs shall be formed using "Gatic" preformed units installed in accordance with the manufacturer's instructions.
 - iii Do not cut any holes, openings, and chases or otherwise interfere with the work of other trades without the specific approval of the. Any damage done to other trades shall be made good without extra charge.
- c Cutting of Concrete Floor Slabs and concrete reinstatement:
 - i Set out lines of trenches and gain structural engineers approval before cutting slabs. Cut slabs to a minimum depth of 100 mm. Break out slabs and remove spoil from site.
 - ii Demolish slabs as required to enable excavation and installation of all hydraulic services.
 - iii Saw cut topping and structural slab in straight lines. The Hydraulic Services Works Package shall backfill and compact trenches etc., as specified. Reinstatement of slab will be by Concreting Works Package.
 - iv The Hydraulic Services Works Package shall remove and dispose of demolished and excavated materials from demolition of topping, slab and trench excavations.
- d Accessibility:
 - i The set out of piping, valves and equipment shall be arranged in conjunction with other trades so as to be readily accessible for operating, servicing, maintaining and repairing of it.
 - ii In no circumstances, shall the Authorities and/or the Contractor conceal any pipes, valves, etc. before inspection and approval.
 - iii Piping, valves, etc. which are installed in unsuitable locations shall be removed and relocated as directed by the. All expense caused by this action shall be made good without extra charge.
 - iv If neither the specification nor plans contain any mention of minor parts, which in the opinion of the Superintendent are reasonably and obviously necessary for the satisfactory completion of the Hydraulic Works Package, such parts are to be provided without extra charge.

16.6 General Piping Installation Requirements

- 1 All screwed joints shall be made with quality pipe joint compound, carefully placed on the threads of the pipe and not through the fittings.

After cutting, all piping, burrs and sharp edges shall be reamed out.

Care should be taken when erecting piping; any marked, dented or otherwise damaged piping shall be replaced in a satisfactory manner.

Piping shall be installed as direct as possible between connecting points and parallel to walls.

Unless otherwise noted, all piping shall be kept as high as possible to beams, slabs, so as to maintain maximum headroom.

Obtain approved ceiling heights and install work to clear obstructions and co-ordinate with work of other trades.

Connections between pipe, fittings, hangers and equipment of dissimilar metals shall be avoided where possible.

Provide a minimum 3 mm thick extruded flexible PVC insulation between supports and service pipe of dissimilar materials and PVC piping. TAPE WILL NOT BE ACCEPTABLE.

Piping shall be carefully installed to provide for proper alignment, slope and expansion, guided and supported in such a manner so that piping will not sag, buckle or cause water hammer.

Provide unions or flanges at connections to each piece of equipment, etc. for easy dismantling and at such other points to facilitate installation.

All fittings and connections at pumps, tanks or other major equipment, 65 mm and over shall be assembled with flanged joints.

All unnecessary lengths of male threads located in exposed pipe work within exposed areas/rooms shall be trimmed to the minimum necessary lengths.

They shall allow for all necessary trimming and machining of threads to obtain the requirements of this clause.

Provide special (and approved) threaded sections where necessary.

Remove all unnecessary hemp or other jointing materials from joints.

Except where otherwise specified, mitred joints shall not be permitted.

The arrangement of all piping shall be generally as shown on the drawings.

16.7 Support and Fixing of Pipework

1 Pipes shall be adequately supported and secured in an approved manner.

Pipes shall not be fixed to, supported by, or welded to other pipes.

Pipe work must be free to move without causing stresses in the pipe work or pipe joints. Where provision has been made for movement in mains, the branch lines shall be unrestrained and in the case of copper tube, annealed for a minimum of 1500 mm from the main. Where this is not achievable, some other approved provision for movement shall be made.

Support all pipes except sheet metal gutters and downpipes with concrete inserts, channels, and pipe clamps, and pipe hangers, etc. in accordance with the manufacturer's instructions and complying with the catalogue regarding spans and loads.

Power-driven fixings, wooden plugs or screw-ins shall NOT be used.

Where pipework supports are installed for water and drainage services within server and comms rooms, supports are to be stainless steel or equivalent of a non-galvanized finish to eliminate 'zinc whisker' formation.

16.8 Building Expansion Joints

1 Crossing building expansion joints with pipelines shall be avoided where possible. Where this is not possible, lines adjacent to a building expansion joint shall be located in a readily accessible position and fitted with an approved expansion joint.

16.9 Joints in Copper Piping

- 1 All joints in copper pipe work shall be made by brazing with low temperature silver brazing alloy containing not less than 5% silver (15% for gas piping). Use oxy-acetylene heating for all low temperature brazing.

Flux shall be as recommended by brazing alloy manufacturer.

Slip joints shall be permitted to join lengths of copper pipes and shall be made by annealing and expanding ends with a proper tool to form a slip (capillary) joint.

Fabricated tees or junctions in copper pipe will not be permitted.

Valves shall not be brazed direct onto pipes. Valves shall be indirectly joined to pipes by the use of flanged or screwed adaptors, which shall be silver brazed onto the pipe and then screwed to the valve. Care shall be taken to prevent distortion to the valve and/or valve seat

16.10 Joints in Copper Piping – press fit

16.10.1 Water

- 1 Connection with press-fit fitting made of copper or gunmetal. Inspection of un-pressed fittings utilises a positive leak path for the whole range 2.2kPa to 300kPa when a dry pressure test is performed and 100kPa to 650kPa with a wet pressure test. This warranted safety function is facilitated by a channel in the metallic body of the fitting. After pressing, the connection is permanent and inseparable with the safety function effectively removed during the pressing process. The seal is made of EPDM, with performance characteristics allowing use in solar applications. The fitting body has a cylindrical pipe guide in front of the seal. Fittings carry Watermark Approval.

16.10.2 Gas

- 1 Connection with press-fit fittings made of copper or gunmetal. Inspection of un-pressed fittings utilises a positive leak path in the whole range 2.2kPa to 300kPa with a dry pressure test. This warranted safety function is facilitated by a channel in the metallic body of the fitting. After pressing, the connection is permanent and inseparable with the safety function effectively removed during the pressing process. The seal is made of HNBR and the fitting body has a cylindrical pipe guide in front of the seal. Fittings carry conformance certifications to international Gas standards DVGW VP 614 and ANSI LC4.
-

16.11 Chasing and Drilling

- 1 Chasing shall be carried out as follows:
 - a Brickwork shall have been built a minimum of 7 days prior to chasing.
 - b Chasing shall be carried out using a machine specifically designed for the purpose. Do not use hand tools.
 - c Chasing shall be on the same side of the wall and vertically above the outlet it serves.
 - d Minimise horizontal chasing.
 - e No chase shall be deeper than 35 mm or wider than 80 mm or 3 conduit pipes. Pipes shall be installed so they are approximately 5 mm below the surface of the bricks. Pipe sizes larger than 25 mm shall be provided with block outs in walls.
 - f Where two or more chases are necessary in close proximity they shall be at least 100 mm apart.
 - g No chases on opposite sides of a wall shall be within 150 mm of each other.
-

16.12 Concealing

- 1 Generally, all piping is to be concealed by chasing or building-in and generally as indicated on the drawings.

16.13 Grades

1 All grades and inverts indicated on the drawings shall be established with the use of approved surveying instruments.

Check the topography and finished floor elevations before commencing any excavation work.

The setting of all inverts, etc. at correct elevations is a prime and definite requirement of this specification.

Grades to drains shall generally be as shown on the drawings and as required by the relevant Authorities.

All water piping shall be graded to facilitate draining at a low-level drainage point.

16.14 Cover Plates

1 Where exposed pipes pass through a floor, wall or ceiling, the joint shall be finished with an approved Chrome-Plated spun metal cover plate. The cover plate is to be flush with the wall.

16.15 Capping Off

1 Throughout the installation of all piping, special care shall be continuously exercised to protect all openings to prevent the admission of dirt, stones or other foreign substances that would subsequently obstruct the system.

16.16 Chrome Plating

1 Waste pipes, and flush pipes, between walls and individual fixtures where in plain sight, shall be chrome-plated and fitted with approved cover plates at walls.

Chrome plating shall be carried out in the best manner.

16.17 Flow Measuring Devices

1 Common manufacture and type throughout project.

1 Low pressure loss type.

2 Magnetic or ultrasonic tube type.

3 Install in accordance with manufacturers' recommendations and with adequate lengths of straight pipe on either side of the device.

4 Label with brass plate stamped to indicate all relevant piping data, gauge ranges, flow and temperature coefficient.

5 Select gauge ranges suitable for each device with the design flow at midscale.

16.18 Gauges and Instrument Fittings

1 Universal Test Points:

- a Binder Twinlock type.
- b Test points to be installed a minimum of 450 above the vertical downwards plane and as required to provide access.

2 Thermometers:

- a 100mm dial type mercury in steel thermometers matching pressure gauges and calibrated in Deg. C.
- b Dial range twice operating temperature range.
- c Accuracy 0.1% of scale range.
- d Bulb type 316 stainless steel.

3 Thermometer Pockets:

- a Locate thermometer pockets in positions where no adverse flow conditions exist.
- b Use thermometer pockets of removable screwed stainless steel type in steel lines, inserted through screwed type boss welded to the pipe.
- c Use thermometer pockets of fixed copper tube type brazed into copper piping lines.
- d Length of pockets and depth of insertion in pipe adequate to ensure accurate readings.
- e Fill pockets with heat conducting fluid to a minimum depth of 50mm and fit caps to each pocket.
- f Enlarge pipes smaller than 65mm so as to retain free area.
- g Install pockets vertically or at an angle to retain heat conducting fluid.

4 Pressure Gauges:

- a Oil filled 100mm dial Bourdon pipe type to AS 1349–1986.
- b Fit with ball valve.
- c Manually adjustable set point needle incorporated on face of gauge.
- d Chrome plated escutcheon and suitable for recessed mounting.
- e Select compound type gauges for pump suctions graduated in mm of mercury on vacuum side and kPa on the pressure side.
- f Bourdon pipe material stainless steel for hot water and steam, phosphor bronze for other services.
- g Bosses for pressure gauges, standard internally screwed type welded or brazed to side of piping.

5 Gauge Panels:

- a Folded sheet metal of adequate stiffness, painted or powder coated with gauges logically grouped and mounted to approval.

16.19 Installation

16.19.1 Arrangement

- 1 Layout piping with adequate provision for expansion, contraction, grading, alignment and access for maintenance with not less than 50mm clearance (including insulation) between pipes and all other surfaces and services.
- 2 Co-ordinate and detail offsets and bends required for co-ordination between trades and integration with building structure and architectural finishes.

16.19.2 Pipework supports and anchors

1 General Requirements:

- a Design supports for all static and dynamic load and reactions.
- b Install seismic bracing to all pipework as required by AS 1170.4–1993.
- c Comply with all Statutory and Code requirements.
- d Locate valves and fittings as close as possible to points of support and separately support valves in line sizes 200mm and larger.

2 Pipe Clamps, Spacers and Saddles:

- a Individually hung pipes – bolted, two-piece type clamps with clevis or welded yokes. One-piece type clamps with clevis may be used for pipe sizes DN65 or less. Clamps rated for the full load of pipe plus its fluid contents, insulation and fittings.
- b Spacers – High temperature insulated pipes - fully vapour sealed hardwood spacers located between clamps and pipes. Spacers wedge shaped hardwood blocks arranged to form circular collars of the same thickness as the insulation and projecting a minimum of 10mm on either side of the clamps.
- c Firmly attach clamps and collars to pipework and/or insulating spacers without applying undue pressure to the pipe walls.

3 Hanger Rods:

- a Fabricate hanger rods and associated fittings from hot dip galvanised or zinc plated mild steel bar of not less than the following diameters:

NOMINAL PIPE SIZE MM	ROD DIAMETER MM
Up to 50 (inclusive)	8
65 & 80	10
100 to 150	12
200 to 250	16
300 to 450	20
500 & above	24

4 Fixed Supports:

- a Fix hanger rods direct to structure unless moving supports required to cater for expansion, contraction and vibration.
- b Support two or more pipes on common channel section bracket or trapeze type support:

PIPE SIZES UP TO 100MM	MILLSTRUT P1000 SERIES
Pipe sizes over 100mm	Standard rolled sections

5 Moving Supports:

- a Where hanger rods used to take up movement ratio of effective length to pipe movement to be not less than 6:1.
- b Provide 'V' shaped brackets with oversize holes for rods. Similarly, clevis or welded yokes on pipe clamps to have oversize holes for rods. Secure rods at each end using spherical washers and locknuts and do not use any nuts between brackets and clevis or welded yoke.

6 Anchors:

- a Design, supply and install anchors where shown on drawings and where required to accommodate pipework weight, thrust forces and expansion and contraction.
- b Submit detailed design and installation details for the project structural engineer's approval and confirmation that the building structure can resist the anchor forces imposed by the system design.
- c Anchor connections to the pipework shall be:

STEEL PIPE:	FULLY WELDED TO PIPE.
Copper pipe:	Anchors comprising bronze flanges fully brazed to pipe and bolted to galvanised steel anchor plates. Use non-conductive bushes and washers.

- d Anchor shall comprise steel plates and/or rolled steel sections designed to transfer the weight of pipework, pipe and thrust forces exerted by the system to the structure.

16.19.3 *Vibration isolators*

- 1 Rubber bellows are not to be used to absorb piping movement due to expansion in the system.
- 2 Rubber bellows should NOT be installed extended, compressed or offset to make up for deficiencies in pipe length or alignment.
- 3 Pipework must be properly anchored around the unit, to avoid the bellows bearing the weight of the piping system.
- 4 Units must be rated for the system operation pressure and temperature.
- 5 The Contractor should note the allowable bellows movement in their submission of products.
- 6 Construction to include flanged connections, bellows to be nylon cord and wire reinforced EPDM rubber suitable for a working temperature of up to 95°C.

16.19.4 *Pipe expansion and contraction*

- 1 Provide means to accommodate piping expansion and contraction in pipe runs longer than 30m from anchor points which may be a series of full radius bends and diversions, expansion loops or bellows.

16.19.5 *Pipe penetrations*

- 1 Supply sleeves for placement and casting in penetrations. Sleeves to extend a minimum of 100mm above floors.
- 2 Pipe penetrations in fire rated structures to comply with BCA Clause C3.15 Fire Resistance – Openings for service installations where applicable – refer to architectural and structural drawings for required fire ratings.
- 3 Pipe penetration fire rating system to comply with approved type tested installation.
- 4 Where required provide 100mm guard around fire rated penetrations to ensure no combustible material comes within 100mm of pipe penetration and pipe.
- 5 Non-fire rated wall penetrations to be sealed with mineral fibre insulation to maintain acoustic rating.
- 6 Where packing material is mineral fibre material enclose in isolation.
- 7 Where exposed to view provide galvanised sheet metal collar.

17 Materials

17.1 Considerations for Materials Engineering

Consideration for proposed materials, used to convey, support and fix services are an essential part of the engineering design process for this project. Materials used for conveying of services could possibly be subjected to accelerated surface corrosion from the sea water spray, high humidity and exposure of Ultra-Violet radiation, if inappropriate materials and or surface treatments are selected.

All materials selected & installed for this project will be suitable for their location and environment. All materials used for conveying of services will have a minimum replacement (life) period of 20 years. All materials used for supporting & fixing of service materials will have a minimum replacement (life) period of 15 years.

17.2 Materials and Workmanship

All materials are to be new and the best of their respective kind. All labour shall be skilled in its particular craft and all shall be in accordance with best practices for works of this kind. The items shall include all labour, equipment, materials and transportation necessary to construct and complete the work.

All pipes and fittings must carry an Australian Standards approval, and watermark approval numbers.

All materials and workmanship not of the stated quality may be rejected.

17.3 Covers and Frames

Covers and frames shall be manufactured in accordance with AS3996 cast iron covers and frames with all edges machine fitted and have removable plastic lifting hole plugs. Each cover shall have a brass or stainless steel plate bearing the name of the service.

The covers and frames shall be set to the level of the finished surface and filled in with the same material as used for the surrounding path or roadways, lawned areas shall be infilled with concrete. Brass edge trimmed covers and frames shall be used in areas where tile, paver or carpet finish is required. All covers and frames shall have the manufacturers recommended lubricant applied prior to installation.

All internal covers shall be lock down type.

Cover duties selected may not be appropriate for construction type traffic. Allow protecting installed covers during the construction phase. All covers damaged shall be replaced without incurring additional costs to the contract.

17.4 Trench Grates and Sumps

All trench grates, roof sumps and outlets, sump grates and frames used, shall be manufactured in accordance with AS3996 and be of the required strength to carry the vehicular or pedestrian traffic. The trench grates used shall be longitudinal bar gratings suitable for vehicle and cyclist traffic, set flush with surrounding finished levels. Trench grates in pedestrian areas shall be of the “heelguard” type and “anti-slip”.

Grates and frames may not be of appropriate duties to withstand construction traffic. Allow to protect installed grates and frames during the construction phase. All grates damaged shall be replaced without incurring additional costs to the contract.

17.5 Fixings

Fixings shall be raw expansion type and shall be installed in direct accordance with the manufacturer's instructions. Plastic or nylon type fixing or anchors will not be accepted.

Sure, drive metal expansion anchors will be accepted on small-bore pipes up to 20mm diameter.

17.6 Flexible Connections

Flexible connections or anti-vibration couplings shall be fitted to all pumping equipment and be 'S' flex flexible couplings installed in accordance with the manufacturer's instructions. Flexible connections shall be the same size as the pipe work, not the equipment.

Flexible connections shall be installed with the aid of adjustable threaded spacers and shall be adjusted correctly to dimensions recommended by the manufacturer prior to fixing of the pipe work, the maximum angular deflection allowed on the flexible connection shall not exceed five (5) degrees and shall be installed only where indicated on the drawings.

Flexible connections on submersible pumps shall be jointed to pump/s and pipe work with male and female 'Kamlok' quick release 'cam type' couplings.

17.7 Flanges

Flanges shall conform to AS 2129, AS 2528 and AS 4087. Use brass flanges for copper tubing, galvanised mild steel flanges screwed type for galvanised mild steel, cast iron flanges for cast iron pipes and PVC flanges for PVC piping.

17.8 Galvanising

All galvanising of steel work shall be hot dip process in accordance with AS 1650 to give 0.1mm minimum thickness coating.

17.9 Unions

Unions shall be four (4) piece compression type unions (2 part cones) or three (3) piece brass, bull nose taper type unions. Brass and copper type union connections shall not be used under any circumstances, in concealed or inaccessible locations.

Unions shall be located on the outlet side of all control valves.

17.10 Valves

All valves used shall be in accordance with AS1628. All valves shall be of one type and of one manufacturer. All valves shall be placed in easily accessible positions for operation and repair. Valves in ducts shall be positioned at 600mm from the finished floor level, unless noted otherwise on the drawings.

Control valves to main branch lines and outlet positions shall be of the gate or ball valve pattern. Elsewhere the control valves shall be of the loose jumper pattern where required by code. All valves shall be tested to a pressure of 1700 kPa and marked by an approved Testing Authority.

Valves up to and including 50mm shall be screwed. Valves above that size shall be flanged with Table E flanges. All valves shall be suitable for temperatures up to 90°C.

Spindles shall be non-rising type and must not project into the bore of the valve when the valve is in the fully open position. The bore must be clear and unobstructed when in this position.

17.11 Reflux Valves

Reflux valves to remain accessible for maintenance purposes. When installed in ground shall have an access shaft to permit replacement of internal components or be located within a full access pit. Valve to be certified for use within drainage system to which it is installed.

17.12 Balancing Valves

Balancing valves shall be self regulating/adjusting type with screwed connections.

17.13 Pressure Limiting Valves

Strainers shall be fitted to the inlet side of the valves. Pressure limiting valves shall be installed in parallel and under no flow conditions, still maintain the desired outlet pressure. The valves and strainers shall be suitable for operation under the pressures and flows it shall be subjected to. The valves shall be adjustable and the outlet and inlet ranges to be approved by the Consultant Engineer.

17.14 Ball Valves

Ball valves shall be manufactured in accordance with AS1628, be fully dezincification resistant, watermarked to MP52, full bore, and screwed BSP connection.

Ball valves may be used in preference to loose jumper type valves where approved by the authority.

17.15 Thermostatic Mixing Valves

The valve must have a wax element which gives complete fail-safe and have built in stop valves for serviceability. The valve shall be a pressure balanced thermostatic mixer.

All valves shall have their temperature set to code requirements and tested following completion of works and all test results shall be recorded and included within the Instruction and Maintenance Manual. Testing shall include failsafe testing with isolation of cold-water service. Provide signage where valves are concealed.

17.16 Sluice Valves

All sluice valves shall be manufactured in accordance with AS2638 and be fully approved.

17.17 Strainers

‘Y’ strains shall be fitted upstream of all back-flow prevention devices, solenoid and similar valves. ‘Y’ strainers shall be approved by the local authority.

17.18 Copper and Brass Tubes And Fittings

Copper tubes shall be solid drawn in accordance with the following standards: -

- 1 Water Service Pipe AS 1432 Type A and B.
-

17.19 Stainless Steel Pipes and Fittings

Stainless steel pipework used for high pressure water services shall be schedule 10S, 316. Jointing method to be stainless steel compatible roll groove.

17.20 Galvanised Steel Pipes and Fittings

Galvanised pipe and fittings to be “heavy grade” for pipework subject to working pressures less than 1700kpa and schedule 40 for pressures exceeding 1700kpa, with roll groove or flanged fittings

17.21 PVC Pipes and Fittings

Sanitary and stormwater drainage shall be unplasticised polyvinyl chloride (PVC-U) drain waste and vent (DWV) with solvent welded jointing. Class of pipe to be equivalent of SH for above ground and in ground up to 3000mm depth, for depths greater than 3000mm class SHE or equivalent.

When specified, water services in ground 100mm and larger shall be PVC-M with a class pressure rating to suit the application with rubber ring jointing. Fittings shall be ductile iron suitable for contact with potable water systems complying with AS/NZS4020.

Where a ring seal adaptor is fitted to the socket of the fittings, care must be taken to leave the required gap between the end of the pipe and the shoulder of the socket of the fittings.

17.22 PE Pipes and Fittings

All pipes and fittings must carry an Australian Standards approval, and watermark approval numbers.

All high density polyethylene (HDPE) pipe work and fitting joints shall be butt-welded or fused, utilising approved butt fusion welding equipment under factory controlled conditions. All site welds may be either butt or fusion, utilising approved butt fusion welding equipment or electro fusion sleeves installed strictly in accordance with the manufacturer's direction. No other jointing methods are to be used.

Pipework used for pressure applications shall be PE100, SDR13.6, PN12.5 at a minimum. Pressure rating shall be increased where required to suit the application.

All persons carrying out the jointing procedure shall complete the installation course and be accredited by the manufacturer prior to commencing works.

Confirmation of accreditation numbers of all personnel intending to install the HDPE system shall be provided prior to commencement of works.

17.23 PE-X Pipes and Fittings

Cross linked polyethylene pipes and fittings must carry an Australian Standards approval, and watermark approval numbers.

Jointing shall be by specifically designed crimp press fit fittings. Push-fit type “pipe grip” fittings will not be permitted. All fittings shall be of same manufacturer of pipework. Crimp tooling shall be approved by the manufacturer of the pipe and fittings. Crimping methods shall be in accordance with manufacturers techniques. Crimp fittings shall not reduce the internal diameter of the pipe, and the selected system must maintain internal bore sizes by way of expanding pipe prior to inserting fittings for crimping.

Where possible pipework colour shall be selected dependant on the type of service.

17.24 PPR Pipes and Fittings

Polypropylene pipes and fittings to be used for chemical sensitive drainage and be rubber ring jointed in accordance with AS 1646.

Large stormwater pipework to be twin walled corrugated polypropylene with rubber ring jointing.

17.25 Polybutylene Hot and Cold Water Pipework

Polybutylene pipes and fittings for hot and cold water shall be a manifold system (as detailed on the hydraulic services drawings) with an individual pipe run from the outlet of the manifold and/or control valve to each fixture and/or shop tenancy.

Jointing shall be press-fit with continuous pipe runs from manifold to fixture, no joints or branches shall be permitted between these two points.

17.26 Jointing

All joints shall be cleaned free of any foreign material, dirt, and grease etc., before any attempt is made to do the joints.

Joints shall be cut true and square, remove all burrs and swarfs, etc. and grind or file a chamfer lightly around the pipe.

Polyvinyl chloride (PVC or PVC-U) pipes jointed by solvent welding of the type recommended by the manufacturer. Clean joints with approved solvent cleaning fluid.

Silver Brazing Alloy shall contain not less than 15% silver; verification of the silver content shall be given in writing to the Consultant Engineer. Care shall be taken not to overheat the joint. Apply the correct flux as recommended by the manufacturer slightly carbonising flame shall be used. Failure to use 15% silver brazing alloy will result in rectification of all joints.

Steel Pipe Welding (Fusion Welding). The joint shall be properly prepared before commencement and will be made in accordance with AS CB15.

Bronze Welding the joint shall be chamfered and the filler rod and flux shall be of the type recommended by the manufacturer.

Sewer Drainage - Rubber Ring Joints shall be in accordance with AS 1646 and installed in accordance with AS 1260.4, AS 2032. All rubber rings shall be of the type recommended by the manufacturer and approved by the local Authority.

Rubber ring jointed in accordance with the AS 1646 and the local Authority, and to be of the type and size recommended by the manufacturer. Where the pipe connecting is of a different material - a varying size ring as supplied by the manufacturer shall be used.

Flange type gaskets shall be material canvas or fibre reinforced rubber of 3 mm minimum thickness.

Lubricant for lubricating the rubber ring and insertion type gaskets and the outside surfaces of the spigot ends or pipes shall be of an approved non-toxic vegetable based type and shall be applied in accordance with the manufacturer's instructions.

Nuts and Bolts generally shall conform to AS 2451 and be heavily galvanised by the hot dip process. Where galvanised iron to cast iron, and cast iron to cast flanges butt, brass bolts shall be used.

Silicone shall have low VOC content and used as recommended by the manufacturer.

17.27 Expansion

Expansion for pressure type services where expansion shall be allowed for because of either excessively long runs exposed to temperature variations (i.e. in excess of 20m), shall be allowed for by means of expansion loops or approved mechanical expansion device.

They shall be of the type, material and manufacture approved by the Authority governing the works or meeting applicable certification level where required, and installed in such a manner that the axial movement of adjacent sections of the pipe and to the extent that the pipe work is above to extend to its possible maximum expansion in both directions without any restrictions.

Expansion joints shall be installed at intervals of no more than 20m and between fixed points exceeding three metres (3m) on all horizontal lines and branches that experience temperature variations and pipes crossing structural expansion joints.

17.28 Temperature Gauges

Provide 100mm diameter waterproof temperature gauges on the flow and return lines of a building flow and return pipe system. Gauges shall be installed to ensure true and accurate readings of adjacent contacting water.

17.29 Pressure Gauges

The face on the gauge shall be minimum of 100mm diameter and gauges shall be waterproof and oriented so that the gauge can be read at a height of 1600mm above floor level.

Each gauge shall be complete with approved gunmetal stopcock and copper piping connected to the relevant pipe work. Provide pressure gauges upstream and downstream of all pumping or pressure reduction devices and plant.

17.30 Thermal Insulation Materials

Pipes shall be lagged to the manufacturer's standards or as specified in hot water insulation section. Where joints, fittings, bends, branches and valves etc., the lagging shall be cut and trimmed to fit around the section and then taped over.

Insulation shall be close celled polyethylene. Insulations shall have no Ozone Depleting Potential (ODP) in either its composition or its manufacture.

Insulation exposed to the elements shall be sheathed in an appropriate material to combat UV and rainwater damage to the insulation material.

17.31 Pressure Switches

Shall be adjusted in strict accordance with the manufacturer's instructions.

17.32 Acoustic Treatment to Pipework

Acoustic treatment, where necessary, shall be supplied and fixed as required to comply with BCA standards, and the acoustic report. Acoustic insulation shown on drawings is indicative only and does not alleviate the contractors responsibility to insulate all pipework as required.

17.33 Protective Pipe Wrapping/Coating

All in ground metal pipes 80mm diameter and greater shall be protected by a polyethylene continuous sleeve (green sock) sealed with a waterproof duct tape. The sleeving shall be fitted to affect a smooth, closely contoured envelope with minimum barrel and socket gaps. There should be sufficient slack in contouring to allow the sleeving to follow the profile of the spigot-socket interface to prevent damage during backfill.

All in ground metal pipes smaller than 80mm diameter shall be wrapped with a layer of petrolatum impregnated fully synthetic woven carrier tape.

All clean pipes and fittings shall be primed with a compatible priming paste prior to wrapping of the tape.

All pipes and fittings shall be covered with a minimum overlap of 30 millimetres.

17.34 Roof Penetration Flashings

All service pipes penetrating the roof shall be made watertight by use of a flexible pipe flashing prior to fitting a cowl.

The pipe flashing shall be designed and installed to conform closely to the roof profile.

Seal the flashing by use of a neutral cure silicone sealant prior to fixing the flashing with 10-gauge zinc plated self-drilling steel screws.

Pipe flashings shall be manufactured from ethylene propylene diene monomer polymer (E.P.M.D.).

17.35 Cold Water Meters

Water meters for domestic services shall comply with local authority requirements including method of flow measurement, and method of authority meter reading requirements. This may include electromagnetic and ultrasonic.

Domestic cold-water services shall be fitted with a low friction turbine meter with integral line strainer, modular meter insert and non-return valve.

All meters shall be fitted with upstream and downstream unions and an upstream control valve, and comply in total with AS 3565.

All meters shall be fitted with upstream and downstream flanges and an upstream control valve.

All meters shall have provision for pulse output to future connection to BMS system. Pulse output shall be 1 pulse per 10 litres.

17.36 Valve Boxes

Shall be cast iron with an identification mark stamped on top.

17.37 Concrete Thrust Blocks

Concrete thrust blocks shall be constructed at all stop ends, fittings, changes of direction and on every fourth straight and continuous length of pipe where required by the pipe material and relevant Australian Standards.

Thrust blocks shall be constructed for a minimum length of 500mm along the pipe and 300mm over the barrel of the pipe, the trench shall be undercut for a depth of 150mm into both sides and the bottom of the trench, reinforcing steel F82 shall be installed to tie the concrete around the pipe and into the rebates at the sides of the trenches.

The thrust blocks shall be supported or backed during the pour with approved formwork and the quantity of concrete shall be not less than 0.5 cubic metres, concrete to be as previously specified, all joints in the pipe work shall be left exposed.

17.38 Backflow Prevention Valves

Supply and install backflow prevention valves in accordance with AS/NZS 3500. All valves shall comply with AS 2845.1. Allow all costs to register each valve with the local regulatory Authority and provide certification.

18 Noise and Vibration

18.1 General Requirements

- 1 Acoustically seal pipe and cable penetrations through noise rated walls and acoustic enclosures.
 - 2 Statically and dynamically balance rotating machinery.
 - 3 Support rotating and reciprocating equipment on anti-vibration/vibration absorbing mounts on hangers.
 - 4 All static equipment in plantrooms with suspended floors to be separated from the structure with double thickness neoprene pads or equal.
 - 5 All connections to rotating equipment or assemblies containing rotating equipment shall be rendered flexible by anti-vibration hangers supporting pipework and/or by flexible connections as specified.
 - 6 Plinths for all floor mounted equipment.
-

18.2 Vibration Control

18.2.1 *Acceptance criteria*

- 1 Complete systems including individual components not to exceed satisfactory levels as set out in AS 1359 and AS 2625.
- 2 Select anti-vibration mounts/hangers for life cycle of building and install in accordance with the requirements of the Earthquake Code AS 1170.4–1993.
- 3 Select and install anti-vibration mounts and hangers in locations to prevent transmission of vibration to adjoining areas and to prevent undue vibration, distortion or fatigue of equipment and connecting piping and cabling.
- 4 Select vibration isolation mounts/hangers to allow for deflections of building structures and finishes and for minimum static deflection as specified.

18.2.2 *Spring mounts*

- 1 Free standing and positioned for uniform deflection of all springs.
- 2 Spring mounting plates and frames hot dip galvanised.
- 3 Ratio of mean coil diameter to compressed length at specified static deflection of 0.8.
- 4 Minimum additional travel to solid of 50%.
- 5 Holding down bolts to be provided.
- 6 Surge frequency to be beyond 30% of predominant machine frequency of vibration.

18.2.3 *Neoprene pads*

- 1 Use up to 9mm static deflection.
- 2 Loading not greater than 400kPa.

18.2.4 *Anti-vibration hangers*

- 1 Spring and double neoprene cup type in series.
 - 2 Total hanger deflection 25mm minimum.
 - 3 Provide secondary adjustment to transfer load to spring.
-

18.3 Noise Measurements

- 1 Measure and submit noise measurements in areas nominated above using sound level meter complying with AS 1259 during the noisiest operating condition.
 - 2 Where noise measurements exceed specified criteria, take additional noise readings in each octave band (125Hz to 4000Hz inclusive) in selected locations with reference to AS 1045-1988, AS 1055, AS 1081, AS 1089, AS 1217 and AS 1469-1983.
-

18.4 Vibration Measurements

- 1 Where a noise hazard exists and all practicable noise attenuation has been carried out, hazardous areas shall be defined by the erection of signs reading:
- 2 Signs shall be 'Brady' Indoor/Outdoor, Personal Hazard Warnings or approved equivalent.

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